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Multiperiod strategic planning of CO_2 capture, transport, utilization and storage with multimodal transportation

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ABSTRACT

To reach decarbonization objectives across the globe, scenario modelling studies indicate that carbon capture, utilization and storage (CCUS) will be essential. Planning CCUS deployment at the regional and national levels requires balancing complex trade-offs – such as when to phase-in infrastructure, where and how much CO_2 to capture, transport, utilize and store, as well as which policy measures to adopt.

We introduce a new multiperiod mixed-integer linear programming (MILP) model for the strategic planning of CCUS, with CO_2 transportation via pipelines, ships, trains and trucks. This model features the selection of capture units and their rates, geographically explicit pipeline network design (including reuse), vehicle fleet estimations, auxiliary transshipment processes, buffer storage, inland reservoirs as well as injection via ships in addition to offshore pipelines. To handle large scale case studies, a simple heuristic algorithm is presented.

The features of the proposed approach are demonstrated on an illustrative Eastern Canada case study involving pipelines, trains and ships. The influence of the temporal resolution chosen by the modeller on the quality of cost estimates and on calculation times is quantified. Compared to leading commercial algorithms, for large model instances the proposed heuristic is shown to produce better results (e.g. 10 % lower cost) with much less computation time (minutes instead of days). This widens the scope of potential use cases, including broader geographical regions and larger sensitivity studies. The annual evolution of an Eastern Canada CCUS value chain is analyzed, paying special attention to the interplay between CO_2 storage reservoir and transport constraints.

1. Introduction

As of 2024, more than 140 countries around the world have committed to reaching net-zero targets in the next decades, representing over 87 % of global GHG emissions (Smit et al., 2024). Achieving these objectives will require a combination of strategies, including enhancing the energy efficiency of current industrial facilities, moving away from fossil fuels via decarbonized energy sources and industrial process changes, electrifying various sectors of economic activity, as well as engaging in carbon capture, utilization and storage (CCUS), as well as carbon dioxide removal. Without CCUS, scenario analyses indicate that decarbonization efforts will likely fall short (IEA, 2017; Lane et al., 2021), namely for hard-to-abate industrial sectors like cement, steel, fertilizer and petrochemicals.

At a high level, a CCUS value chain involves capturing CO_2 emissions at industrial sites across different sectors, such as cement, steel, pulp and paper, oil and gas, bioenergy, fertilizer and chemicals, combustion-based heat and power, as well as fossil fuel-based hydrogen generation. The captured emissions are then transported via a mix of pipelines, ships, trains and trucks to onshore or offshore geological sites (e.g. saline aquifers, depleted oil and gas fields) where they are injected below caprock to be stored for 10,000+ years (Alcalde et al., 2018). Although at an expected smaller scale than geological storage, captured CO_2 can also be utilized, e.g. for enhanced oil recovery or to produce fuels and chemicals.

In 2021, Canada affirmed through legislation its commitment to achieve net-zero carbon emissions by 2050 (Canadian Net-Zero Emissions Accountability Act, 2021). Although some high-level preliminary assessments have been made in the scientific literature (Zhang et al., 2023), the strategic planning and deployment of Canadian CCUS infrastructure presents challenges. In particular, provincial and federal policy programs must be based on techno-economic-environmental evidence

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taking into account the heterogeneous landscape of industrial emission sources across the country. Moreover, existing infrastructure and geographical features must be factored in the analysis. This includes diverse geology, topography, waterways, protected lands (wildlife, First Nations), population density, as well as existing infrastructure (e.g. roads, railroads, electrical grid, oil & gas pipelines).

Expanding CCUS infrastructure over time will presumably imply an increase in the CO_2 volumes to be captured, transported and sequestered. On the transportation side, pipelines are often favoured to transport large volumes of CO_2 via trunklines to benefit from economies of scale. However, pipeline development typically takes years to complete and the trade-off between building excess pipeline capacity in advance versus adding capacity gradually (e.g. via parallel lines) must be evaluated. For new pipelines, CO_2 can be transported in supercritical phase due to its economically desirable fluid properties (e.g. high density, low viscosity) and acceptable corrosion rates (IPCC, 2005). However, this mode of transport requires rigorous safety assessments and compliance with evolving regulatory standards, particularly in or near populated areas. Based on a recent inquiry of pipeline operators (Cauchois et al., 2021), gas phase CO_2 may be favoured for CO_2 transportation via repurposed oil and gas pipelines within safe operating pressure ranges.

Vehicle transportation of liquid CO_2 on railcars, ships and trucks offers a means of moving smaller volumes. In particular, vehicles can be used to adapt to the ramp-up phase of transport volumes while pipelines are being built, or when pipeline economies of scale are insufficient and pressurization costs are prohibitive. Vehicles can also enhance network robustness by providing a backup transportation means in the event of temporary pipeline link failures, e.g. unplanned maintenance (Gabrielli et al., 2022). They can be a viable alternative to pipelines depending on existing transport infrastructure (e.g. railways, roads) as well as on the distances and volumes involved. In particular, for inland emitters located far from existing or planned pipeline trunklines, existing rail networks can offer a cost-effective substitute to pipelines when only small railway extensions (spurs) are necessary to reach the appropriate sites. Compared to rail transportation, trucks can cover shorter distances and transport smaller volumes. For transportation to offshore injection sites or to intermediate floating storage platforms (FSI), ships can be an economically viable alternative to offshore pipelines when distances are above a certain threshold (e.g. 500-1000 km), determined by a ship's tank pressure and the flow rate involved (IEAGHG, 2020; Baroudi et al., 2021). Transitioning from liquid phase transport in vehicles to supercritical (or gas) phase pipelines (and vice versa) involves state change processes (e.g. liquefaction, heating-compression, (re)conditioning, depressurization). Since vehicle transport is inherently batchwise and pipeline flow is continuous, transshipment from one to the other requires buffer storage (i.e., temporary tanks). For a comparison of CO_2 transport modes in terms of their technology readiness level, contributing cost factors and other considerations, the reader is referred to Galeazzi et al. (2023).

To be an impactful carbon management measure, the pace of CCUS implementation in Canada and elsewhere must accelerate (Lane et al., 2021). To support this effort, numerical optimization models provide a comprehensive view of the trade-offs involved. These include the influence of policy measures (e.g. carbon pricing, subsidies) on attaining CO_2 capture targets and costs, as well as the impact of risk and uncertainty (e.g. geological uncertainty, environmental risk), the latter which is not covered in this work.

The present study introduces a new multiperiod mixed-integer linear programming (MILP) model for the strategic planning of CCUS, including transportation by pipelines, ships, trains and trucks. This model constitutes the core of the strategic planning tool in Natural Resources Canada's (NRCan) National CCUS Assessment Framework (NCAF), a decision support platform for the strategic planning of large-scale CCUS projects in Canada (Symonds et al., 2025). It includes geographically explicit pipeline network design, as opposed to direct source-sink matching present in other models. To handle large scale case studies, a simple heuristic algorithm is proposed to lower calculation times while yielding high quality solutions. The features of the proposed model and algorithm are highlighted on an illustrative case study for Eastern Canada.

This paper is organized as follows. The next section provides an overview of the scientific literature on multiperiod strategic planning models for CCUS. We then summarize the scope and highlight the contributions of the present study. In Section 3, the problem addressed is detailed and the mathematical formulation proposed is described. Section 4 describes the proposed heuristic algorithm and Section 5 presents an illustrative Eastern Canada case study. Optimization results are then analyzed. Concluding remarks and avenues for future work are identified in Section 6.

2. Literature overview and contribution

2.1. Literature overview: modelling

Over the years, various optimization models have been developed for the strategic planning of CCUS value chains. For an in-depth review of the topic, the reader is referred to Zhang et al. (2022) and Tapia et al. (2018). For a short overview of national scale ship and pipeline studies, see d'Amore et al. (2021b). For a treatment of pipeline-specific CCUS value chain modelling, see Homsi et al. (2025). We complement these reviews by focusing on models that include multimodal transportation. At a high level, CCUS strategic planning models consider the temporal evolution of decisions in one of two ways, either by considering the time horizon as one static block (single period models), or by decomposing it in a finite set of time periods (multiperiod models). Single period planning models can offshoot costs by more than 50 % (Middleton et al., 2012). As a consequence, this literature overview centers on multiperiod models. Approaches to this problem can be partitioned into two categories: those that are geographically approximate (Han and Lee, 2011a) and those that are geographically explicit (Middleton et al., 2020) in their transportation routing plans. Only a minority of modeling frameworks fall into the latter category. In order to instantiate what will be referred to as geographically explicit models, a candidate pipeline network encapsulating promising pipeline routes is determined before carrying out the numerical optimization. These potential pipeline paths can be generated by first rasterizing the geographical features of the region of interest (Hoover et al., 2020). A cost is then associated to each possible pipeline path by associating a weight factor to each geographical feature, while also considering the orientation of pipeline segments passing through the center of each raster cell. A candidate pipeline network can then be generated by calculating least-cost paths (e.g. via Dijkstra's algorithm) for each source-sink pair, or more efficiently, for each pair of adjacent nodes in a Delaunay triangulation (Middleton and Bielicki, 2009; Morbee et al., 2012). In this work, we designate as "geographically explicit" the approaches that include such candidate pipeline networks as well as precise shipping routes, roads and railroads. Other approaches are termed "geographically approximate".

In the context of CCUS investigations in Korea, the research team at POSTECH proposed multiple optimization models over the years. Han and Lee (2011a) introduced a single period MILP model with pipeline, ship, rail and truck transportation. With no explicit transportation network or routing, this model pairs geographical regions using average distances. The authors apply a superstructure including various types of capture technologies and industries to four geographical regions, three geological storage areas and two utilization regions, each with multiple technological options. This model is extended to a multiperiod context in Han et al. (2012) and tested on a modified version of the case study originally introduced in Han and Lee (2011a) with four 5-year time periods from 2011 to 2030. In Han and Lee (2011b), the authors put forward a 2-stage stochastic programming version of their single period

model (Han and Lee, 2011a) with uncertain CO_2 emissions. Continuing their work on multiperiod planning, the authors introduce a two-stage scenario-based stochastic programming version (Han and Lee, 2012) of their earlier model. Uncertain parameters include emission profiles, costs and CO_2 utilization product prices. In Lee et al. (2017), a new stochastic version of the single period model in Han and Lee (2011a) is proposed in the form of a two-stage stochastic multi-objective optimization model with environmental risk aspects embedded in the eco99 life cycle analysis indicator (Goedkoop and Spriensma, 2001). In Nam et al. (2013), a team at KAIST propose two MILP models for CO_2 shipping in the Korean context, one for liquefaction facility location, the other for fleet management, fleet sizing, fleet mix (i.e. vessel types) and route assignment.

Bennæs et al. (2024) consider CCUS strategic planning involving ship and pipeline transportation. Their case study spans up to 34 German emitter sites for which the port of Wilhelmshaven is used for loading captured emissions to be sent to the Northern Lights port infrastructure for unloading. Offshore pipelines then bring those captured emissions to the Kollsnes offshore reservoir in Norway. The authors analyze four scenarios of total CO_2 captured per annum. In an attempt to ensure operational feasibility, for each scenario the proposed model is solved for 30 days of operations discretized in periods of 12 hours. The pipeline part of the proposed model is not geographically explicit. As a consequence, pipeline economies of scale cannot be appropriately modelled (e.g. trunk lines).

Within the scope of the Grenland hub in Norway, Fraga et al. (2021) propose a geographically approximate multiperiod MILP model for ships and pipelines. In the context of the Northern Lights initiative, Bjerketvedt et al. (2022) put forward a multiperiod MILP model for pipelines and ships, including fleet sizing and fleet mix. This model excludes capture costs and focuses on shipping value chains from conditioning to reconditioning, including intermediate temporary storage tanks. The same group of researchers also propose a 2-stage stochastic programming model for ship transportation, with special attention on uncertain ship speed (or travel times) due to weather conditions (Bjerketvedt et al., 2020). As part of the AlignCCUS EU project, Nie et al. (2021) put forth a two-stage scenario-based stochastic programming model with uncertainty on storage capacity and CO_2 emissions. This multiperiod MILP model includes CO_2 transportation by pipelines, ships and trucks. It is applied to the planning of the UK Grangemouth cluster for a time horizon ranging from 2020 to 2055, discretized in 7 time periods. To define pipeline routes, the authors extend the approach of Zhou et al. (2014) via a Hanan grid. These approaches have the advantage of being able to establish approximate pipeline routing and capacities in a single step. This contrasts with previously termed geographically explicit methods, which first define a network of candidate pipeline routes and then optimize this network to determine actual routes, flows and capacities in a second step (Middleton and Bielicki, 2009; Morbee et al., 2012; Hoover et al., 2020). However, an important hypothesis of the work of Zhou et al. (2014), as acknowledged by the authors themselves, is that geography is uniform, implying that it doesn't impact pipeline costs. While this may be applicable to some regions, it does not apply to geographically diverse countries like Canada and the United States.

Over the last decade, the CAPE-Lab in Italy has introduced various CCUS value chain optimization models. In d'Amore and Bezzo (2017), the authors propose a multiperiod MILP model with ship and pipeline transportation (inland, offshore). They apply it to a Europe-wide CCUS value chain over a 20-year horizon, decomposed in twenty 1-year periods. Given the scope of the region of interest, their model is geographically approximate and partitions Europe in 124 cells. They propose a two-stage numerical resolution method consisting of a continuous relaxation model followed by a reduced MILP. Calculation times recorded can go up to several hours. In d'Amore et al. (2018), a societal risk assessment study quantifies potential health impacts of hazardous incidents related to inland pipelines. In order to ensure computational tractability, the MILP model used consists of a single time period and encompasses mainland Europe (124 cells) and the North Sea offshore area (134 cells). In d'Amore et al. (2019), the model presented in d'Amore and Bezzo (2017) is extended to the context of reservoir uncertainty. A Europe-wide study containing more than 400 emitters from cement, steel, refining and power sectors and 50 sequestration nodes is carried out in d'Amore et al. (2021a). Given the large volumes to be transported, the authors focus only on pipelines (inland andoffshore) and use a single-period MILP model. The scope of this model is broadened in d'Amore et al. (2021b) to include ship transportation for storage in the North Sea.

2.2. Literature overview: calculation times and algorithms

Although CCUS strategic planning spans multiple decades, the practical application of numerical optimization in that context requires the timely generation of quality solutions (Whitman et al., 2022; Homsi et al., 2025). This is partly due to the growing scope of strategic evaluations and the increasing number of factors to be considered. In some countries (e.g. the US), the scope of strategic planning is shifting from regional to country-level considerations involving thousands of prospective emitter and storage sites (Talsma et al., 2022). Moreover, as reservoir characterizations advance, an increasing number of studies will move from modelling reservoir sites as single centroid nodes towards more precise evaluations involving 10-100+ injection locations (Talsma et al., 2022). Even in the context of single period deterministic CCUS strategic planning with pipelines only, MILP solvers can yield large optimality gaps (e.g. 40 %) after days of calculations (Whitman et al., 2022). This issue is exacerbated when accounting for the time-dependent nature of decisions (Homsi et al. 2025). To tighten the integrality gap, classical operations research valid inequalities cannot be directly applied to this problem (Homsi et al., 2025), namely due to not having a fixed demand per sink. The reader can refer to Homsi et al. (2025) for further discussion of this topic. In addition, the use of such models to inform decision-making by government stakeholders may necessitate 100-1000+ model evaluations to quantify the impact of policy measures, such as carbon taxes (Middleton and Brandt, 2013; Ellett et al., 2017), subsidies and capture targets.

These factors will contribute to a significant growth in model size and in the number of runs that are required. To address this problem for pipeline-based CCUS, both Whitman et al. (2022) and Homsi et al. (2025) propose slope scaling heuristic variants, the former in a deterministic single period context and the latter for deterministic multiperiod assessments. In a geographically explicit multimodal transportation context (ships, trains, trucks), to the authors' knowledge, no such algorithm has been proposed so far in the literature.

2.3. Scope and contribution

The present study focuses on *deterministic* multiperiod strategic planning of CCUS with *geographically explicit* multimodal transportation (pipelines, ships, trains and trucks). These features are essential to properly take into account the temporal evolution of infrastructure, as well as the variety of geography and transportation options available in Canada. The modelling envelope considered covers the whole CCUS value chain, i.e. capture, transport, storage and utilization. Section 3 details the strategic planning decisions that define the modelling scope.

In order to situate this work in the scientific landscape, we summarize research gaps stemming from the literature survey presented in Section 2.1 and 2.2. The previous bibliographical review reveals only two main multiperiod models for CCUS strategic planning that include *geographically explicit* transport. These are the models of Middleton et al. (2012) and Jones et al. (2022). Both are deterministic and solely consider transportation via pipelines. The main difference between these models is in how they treat pipeline costing. The former (Middleton et al., 2012) considers pipeline diameters explicitly (nominal pipeline diameters) while the latter (Jones et al., 2022) uses a piecewise

linear approximation of pipeline costs replacing diameters by two "pipeline capacity trends", i.e. intervals of maximal pipeline flow (Middleton, 2013). This piecewise linearization improves the poor results and calculation times observed in pipeline diameter-based models (Middleton, 2013). However, as explained in Section 2.2, even for single-period instances commercial solvers struggle for these piecewise linear pipeline-specific models. This has led to the introduction of heuristics that focus solely on pipeline-based geographically explicit CCUS (Section 2.2). Further work remains to be done in that field as classical operations research methods to strengthen the formulation, e.g. network design valid inequalities, cannot be directly applied to the CCUS context (Homsi et al., 2025).

Some considerations related to CCUS strategic planning are of a stochastic nature. These include geological reservoir uncertainty impacting capacities and injectivities. In addition, storage site availability is subject to uncertain investor confidence, potentially undefined regional regulations (e.g. brine disposal) and competing land use interests such as rare mineral mining (Lane et al., 2021). Additionally, future emission profiles available for capture are also uncertain, due to market changes and industrial process changes (e.g. due to electrification). The previous literature survey revealed no geographically explicit stochastic programming model for CCUS strategic planning. It should be noted that this is likely due to the aforementioned computational challenges that face *deterministic* pipeline-specific models. In particular, going from deterministic to stochastic programming would likely lead to a significant growth in the number of equations and variables, as they would be indexed per scenario and per stochastic programming stage. This increase in problem size would be compounded by an increase in the number of storage nodes due to evaluations moving from geological formation-level assessments to more local injection site assessments (Section 2.2). These factors would likely lead to larger instances, slower linear relaxations at branch and bound nodes and thus poorer overall computational performance. Following these observations, this work focuses on deterministic modelling, with the intention of adding stochastic elements in future model versions.

To address the identified gaps in the published literature, we propose a deterministic multiperiod CCUS strategic planning model with geographically explicit multimodal transportation. This model includes the following novel features.

- For pipeline transportation, we introduce an extension of the geographically explicit CCUS model of Jones et al. (2022) to explicitly distinguish between gas and supercritical pipelines. The new model also adds the option to differentiate between building new pipelines and repurposing existing ones.
- For vehicle transportation, we propose a reformulation of the geographically approximate fleet sizing and fleet mix models of Han and Lee (2011), Fraga et al. (2021) and Nie et al. (2021) in a geographically explicit context with ships, trains and trucks. Contrary to these models, the proposed model includes new equations to explicitly distinguish between using existing vehicles and purchasing new ones. It also includes vehicle cost functions that generalize the ones found in those models. This gives the modeler the flexibility to define those costs in alternative ways depending on the data available and the analysis at hand (Section 4.3.12). We also introduce new equations to extend existing roads and railroads to reach otherwise inaccessible sites and add related costs.
- In contrast to the CCUS models of Han and Lee (2011), Fraga et al. (2021) and Nie et al. (2021), we explicitly model the annual flows and capacities of buffer storage and state change processes, namely those necessary for transshipment in a multimodal transport setting. These include liquefaction, heating-compression, (re)conditioning and depressurization at all value chain nodes. In contrast to those models, we propose equations for the capacity expansion of those processes over time. Moreover, the proposed buffer storage capacity equations generalize the rules of thumb from the IEAGHG (2023). Contrary to these engineering heuristics, these new equations can be applied to any value chain topology.
- For offshore storage, we propose new equations to represent injection via offshore pipelines and via ships, either through floating platforms (FSI) or directly through injection ships. In particular, we introduce a novel mathematical formulation that determines the time ships spend unloading for direct and indirect injection at offshore reservoirs. This allows the evaluation of key trade-offs identified by the IEAGHG (2020) between costs for injection, shipping, offshore pipelines, liquefaction and (re)conditioning. The present literature review revealed no CCUS model with those features.

Since reaching high quality solutions in a timely manner can be a barrier to the use of optimization models, we introduce a simple heuristic algorithm intended for large case studies. The literature review conducted in Section 2.2 identified no such algorithm for the multiperiod planning of CCUS with geographically explicit transport via pipelines, ships, trains and trucks.

3. Problem statement

This section gives a general description of the optimization problem under consideration. Throughout this text, the term "state change process" refers to any process used to bring CO_2 to an acceptable state (pressure, temperature, phase, composition) for buffer storage tanks and transshipment. The term "capture unit" refers to the combination of an emitting process with an associated capture technology, separation and purification processes, as well as final CO_2 compression, drying or liquefaction. Capture units can be designed to output CO_2 either in gas, liquid or supercritical phase.

The optimal CCUS strategic planning problem considered in this study can be stated as follows. Given are the following optimization sets and parameters.

- A set of time periods, possibly of unequal duration, which discretize the planning horizon.
- A set of value chain nodes corresponding to emitters, sinks – i.e. geological reservoirs (inland, offshore) or utilization locations - ports, transshipment hubs and junction nodes.
- A set of candidate pipeline network arcs, including potential routes for new and repurposed CO_2 pipelines (gas or supercritical).
- A set of existing rail and road network arcs.
- A set of arcs representing potential extensions for rail and road networks, respectively.
- A set of potential shipping route arcs.
- Sets of vehicle types for ship, rail and road transportation.
- A set of capture units at each emitter site, each with gas, liquid or supercritical CO_2 output.
- Parameters for each vehicle type: cargo capacity, average speed on each arc, fuel consumption, (un)loading rates, berth time (ships), maximum active hours in a year, initial (i.e. existing) fleet size at the beginning of the time horizon.
- Parameters for reservoirs: injectivity (i.e. maximum injection rate) and operational lifetime capacity. For offshore storage, this includes direct injection (i.e. from ships) and indirect injection (i.e. from intermediate FSI).
- Parameters for capture units: maximal capturable emissions annually.
- Parameters for transshipment at nodes: vehicle (un)loading rate limits at emitters, geological reservoirs (inland, offshore), ports and transshipment hubs.
- Parameters for buffer storage and FSI: safety margins, limits and initial values for buffer storage and FSI capacities and levels.
- Parameters for state change processes: production ratios, i.e. ratio of outlet to inlet CO_2 flows.

- Parameters for costs of all capture, transportation and storage options.
- Parameter for carbon taxes for each emission point (if applicable).
- Parameters for global capture and storage targets for each time period, i.e. cumulative over all capture units and reservoirs, respectively.

The objective function (to minimize) is the total cost for capture, transportation and storage taking into account the following optimization variables.

- Choice of emitter sites, capture units and annual capture rates for each emitter site.
- Choice of inland and offshore geological storage sites and annual injection rates.
- For offshore storage, choice between direct and indirect injection (i.e. via FSI).
- Selection of locations where new pipelines should be built and existing ones should be retrofitted, their capacity and annual flow.
- Selection of transport routes for ships, trains and trucks and their annual fleet flow.
- Selection of road and railroad extensions to build.
- Ship, train and truck fleet sizing (i.e. number of vehicles) and fleet mix (i.e. choice of vehicle types).
- For each arc in the vehicle transportation network, the number of trips of each vehicle fleet traversing that arc and their annual cargo load.
- For each node in the value chain network, annual buffer storage capacities and levels, internal annual flows associated to CO_2 state change processes, as well as vehicle fleet loading and unloading rates.

Carbon dioxide utilization, such as enhanced oil recovery, can be considered via storage nodes with negative cost parameters (Middleton et al., 2011). More detailed utilization use cases are the subject of future research.

4. Model formulation

In this section, we formalize the previously introduced strategic planning problem as a MILP model. We briefly position this model in the operations research landscape and list the underlying assumptions made. Afterwards, the mathematical formulation itself is detailed.

4.1. Model classification

In a broad sense, the proposed model combines features of various operations research problem classes and adapts them to the CCUS context. Geographically explicit pipeline related decisions fall into the network design problem class, while site and technology choices are akin to facility location problems. Vehicle CO_2 transportation management is treated as a mix of fleet sizing (number of ships, trains, trucks) and fleet mix problems (vehicle types). At each network node, the internal CO_2 flows between pipelines, vehicles, storage tanks and CO_2 state change processes are modeled using mass balances and rate equations based on simplified flow diagrams. All these attributes are included in a single model.

4.2. Assumptions

In order to ensure computational tractability and to reflect the fact that detailed operational information is not always known when planning at the strategic level, the following modelling assumptions are made.

- All processes are assumed to operate at a steady state.
- For each time period, all decisions, including annual flows, capture rates and injection rates, are assumed to be identical for each year within that time period. These decisions can (and likely will) change from one time period to another. The user can adjust the duration of each time period as a function of computational performance. Time period duration can also be adjusted to reflect the level of detail required by the case study at hand, e.g. as a function of the confidence in economic and market forecasts.
- Vehicles are assumed to make round trips. This hypothesis is common when establishing fleet sizing and fleet mix, since individual vehicle scheduling is done at the operational (or tactical) level, not when making strategic plans. Combining strategic and tactical aspects would potentially result in an intractable model and would require daily operational information not available at the strategic level (e.g. transportation contracts with carriers).
- Infrastructure build times are not explicitly modeled but can be accounted for implicitly via the choice of time period durations. Infrastructure is assumed to be available and operational from the moment it is selected, i.e. at the beginning of the corresponding time period onwards. This hypothesis is common (Middleton et al., 2012; Jones et al., 2022) and could be eliminated in future versions of the model, with potentially negative implications on computing performance.
- It is assumed that if offshore injection from ships is considered (injection ships or FSI), then either all ships do direct injection, or they all unload into an intermediate FSI for indirect injection. Offshore pipelines can be combined with these options.
- Transport ships cannot directly inject CO_2 on the seabed floor, but can (un)load at ports, as well as unload onto FSI. Injection ships, as their name indicates, can directly inject CO_2 at offshore storage sites (onboard conditioning) as well as (un)load at ports and unload onto FSI.
- All model parameters (capacities, costs, etc.) are deterministic.

4.3. Mathematical formulation

In this section, the detailed formulation of the MILP model is presented. To help the reader understand the forthcoming mathematical equations, we begin with an overview of the notation used. Throughout the model description, when new symbols arise that are not covered in this overview, a brief explanation is given. Appendix A contains all notation details.

In terms of units of measurement, quantities of CO_2 are expressed in megatonnes, time is in years, flows are in megatonnes per year, distance is in kilometers and costs are in millions of dollars. The following letters are used as subscripts or superscripts for sets, variables and parameters to denote various value chain elements: emitters (e); geological reservoir sites (g); inland reservoirs (inl); offshore reservoirs (sea); ports or ships (s); railroads, railcars or trains (r); roads or trucks (t); pipelines (p).

Since the proposed model is *geographically explicit*, a candidate transportation network (i.e. directed graph) containing all potential transport routes is assumed to be given *a priori* (Section 2.1, Section 3). The set of all underlying nodes in this network is denoted by I. It is partitioned into the following subsets: emitter sites (I_e), geological reservoirs (I_g), ports (I_s), transshipment hubs (I_{hub}) and junction nodes (I_{mix}). A transshipment hub is a node solely dedicated to the exchange of CO_2 cargo between transport modes. It includes temporary storage (cryogenic tanks) and CO_2 state change processes (compression, conditioning, etc.). On the other hand, a junction node is transportation mode specific and does not allow transshipment. Conceptually, it consists in a mixer followed by a splitter, e.g. a fork or junction in a pipeline, road or railroad.

Vehicle types are included in the set V, which is decomposed into the set of train (or railcar) types (V_r), ship types (V_s) and truck types (V_t). Ship types are further partitioned into injection ships (V_s^{dir}), which have

direct injection equipment on board and transport ships (V_s^{ind}), which can unload unto an FSI for indirect injection but don't have direct injection equipment on board.

In the proposed model, the use of subscript v (in the set V) refers to the whole fleet of vehicles of type v, as opposed to a single vehicle (see variables explanation below). Instead of considering nominal pipeline diameters, pipeline capacities are modelled via capacity intervals (set C) in which maximum flows can be selected, as introduced by Middleton (2013). See pipelines subsection below for details (Section 4.3.6).

Each arc in the candidate network has an origin and a destination node in the set I. Single arc loops are not permitted, i.e. arcs having the same origin and destination node. The set of all arc labels is represented by K. If vehicles of type v can transport CO_2 along arc k (e.g. if v is a truck type and k is a road) from node i to node j, then the tuple (k,i,j,v) is added to the set N_V. On the other hand, if arc k from node i to node j can admit a supercritical or gas pipeline with capacity trend c, then the tuple (k,i,j,c) is added to the set N_{sup} or N_{gas}, respectively. To specify that vehicles of type v are permitted to load or unload at a site node i, the tuple (i,v) is added to the set L or U ("loading", "unloading"), respectively.

Emitter nodes can contain multiple potential capture units, included in the set M. For each potential capture unit m at an emitter site i, the tuple (i,m) is added to the set I_M. This set is partitioned into I_M^{gas}, I_M^{sup} and I_M^{liq} depending on the phase of a capture unit's output CO_2 stream (gas, supercritical or liquid, respectively).

The proposed model uses binary, integer (non-negative) and continuous variables to describe the CCUS strategic planning problem. When interpreting variables and equations, the reader should keep in mind that during a time period, all yearly decisions are assumed identical (see Section 4.2). Binary variables are symbolized by the letter y, with or without superscripts. The selection of a node i (emitter, port, transshipment hub, reservoir) at the beginning of time period h is denoted by the binary variable $y_{i,h}$, i.e. that node becomes available from that time period onwards if and only if $y_{i,h}$ takes value 1. If an emitter site i is retrofitted with capture unit m at the beginning of time period h, then the binary variable $y^e_{i,m,h}$ takes value 1 (0, otherwise). If a pipeline of capacity interval c is built on candidate network arc k during time period h, then the binary variable $y^p_{k,c,h}$ takes value 1 (0, otherwise). If an offshore reservoir site i starts using indirect injection (i.e. injection via an FSI) at the beginning of period h onwards, then the binary variable $y^{ind}_{i,h}$ takes value 1 (0, otherwise). If that site starts using direct injection (i.e. injection directly from transport ships) at the beginning of period h onwards, then the binary variable $y^{dir}_{i,h}$ takes value 1 (0, otherwise).

Integer variables (non-negative) are symbolized by the letter n, with or without superscripts. The size of the fleet of vehicles of type v used in period h is counted via the integer variable $n_{v,h}$. This fleet includes new vehicles of type v purchased (and available) at the beginning of period h, denoted by $n^{\text{new}}_{v,h}$. When h = 0, $n^{\text{new}}_{v,0}$ corresponds to the number of vehicles available at the beginning of the time horizon. On a transportation arc k (e.g. a road), the number of trips (or "outings") of the fleet of vehicles of type v along arc k during period h is denoted by $n^o_{k,v,h}$.

Non-negative continuous variables are used to describe annual flows, capture rates, inventory levels and other quantities. We list the main ones here; see Appendix A for an exhaustive list. Annual flows, capture rates and injection rates are denoted by x, a and b, respectively. For example, $x_{k,v,h}$ represents the annual quantity of CO_2 transported by the fleet of vehicles of type v on arc k during period h. The annual CO_2 flow in pipeline arc k for pipeline capacity trend c during period h is represented as $x^p_{k,c,h}$; the corresponding pipeline capacity is written as $q_{k,c,h}$. At an emitter site i, the annual capture rate of unit m during period h is denoted by $a_{i,m,h}$. At a reservoir site i, the annual injection rate coming from pipelines during period h is symbolized by $b^p_{i,h}$, while the annual injection rate of CO_2 coming from ships (at offshore sites) is described via $b^s_{i,h}$. The annual buffer storage level at node i during time period h is denoted by $f_{i,h}$. Other notation is introduced as it occurs throughout the model description.

4.3.1. Inland nodes: mass balances

At emitters, inland geological reservoirs, ports and transshipment hubs, Fig. 1 gives a superstructure embedding the CO_2 flows considered. These correspond to:

- vehicle (un)loading;
- CO_2 state change processes;
- temporary storage (buffer tanks);
- capture units (at emitters); and
- injection (at reservoirs).

The dotted line symbolizes the envelope of the node considered. Input and output CO_2 streams for that node correspond to the arrows intersecting the dotted boundary. To reduce visual clutter, multiple capture units and injection points are symbolized by a single box per CO_2 phase, indices on variables for internal flows were dropped (e.g. x_1^{gas} instead of $x^{gas}_{1,i,h}$) and input-output arcs are aggregated by type (e.g. pipelines, ships), as well as injection/capture flows.

To reduce the number of equations, the symbol 1_ξ is used throughout to denote the indicator function, which takes value 1 when logical proposition ξ is true and value 0 otherwise. Mass balances for vehicle loading and unloading (rail, ship, truck) are encapsulated in Eq. 1. For junction nodes (I_{mix}), i.e. mixer-splitter nodes, there is no vehicle loading or unloading. Therefore corresponding terms in this equation vanish (via the indicator function 1).

$$\sum_{(k,j):(k,i,j,v)\in N_V} x_{k,v,h} - \sum_{(k,j):(k,j,i,v)\in N_V} x_{k,v,h} = 1_{(i,v)\in L} x^{\text{load}}_{i,v,h} - 1_{(i,v)\in U} x^{\text{unload}}_{i,v,h}, \quad (1)$$

$$\forall (i,v) \in I_V, h \in H \text{ with } i \notin I_g^{sea}$$

The variables $x^{\text{load}}_{i,v,h}$ and $x^{\text{unload}}_{i,v,h}$ denote the annual loading and unloading rates (respectively) of the fleet of vehicles of type v at node i during time period h. Recall that this model does not track individual vehicles, but rather fleets of vehicles of each type. Therefore, flows as well as loading and unloading rates refer to those of fleets of vehicles.

When inlet and outlet vehicle flows are zero, the left-hand side of Eq. 1 vanishes and the difference $1_{(i,v)\in L} x^{\text{load}}_{i,v,h} - 1_{(i,v)\in U} x^{\text{unload}}_{i,v,h}$ is forced to zero. In that context, the following equation ensures that both loading and unloading rates vanish. To see this, note that when there is no vehicle flow to and from a node, the right-hand term of Eq. 2 is zero, which forces the *non-negative* variable $x^{unload}_{i,v,h}$ to zero. This in turn forces $1_{(i,v)\in L} x^{load}_{i,v,h}$ to zero, since in that case the difference between loading and unloading rates also vanishes (Eq. 1).

$$x^{unload}_{i,v,h} \leq \sum_{(k,j):(k,j,i,v)\in N_V} x_{k,v,h}, \forall (i,v) \in U, h \in H \quad (2)$$

For emitter sites, inland reservoirs, ports and transshipment nodes, material balances for gas and supercritical pipelines are given in Eqs. 3 to 6 (see Fig. 1), while those at junction nodes are given by Eq. 7. Annual buffer storage levels at emitters, inland geological reservoirs, ports and transshipment nodes are defined by Eq. (8). In the Eqs. below, the variables $x^{gas}_{1,i,h}$, $x^{gas}_{2,i,h}$, $x^{liq}_{1,i,h}$, $x^{liq}_{2,i,h}$, $x^{sup}_{1,i,h}$, $x^{sup}_{2,i,h}$ and $x^{sup}_{3,i,h}$ refer to the annual flows of internal CO_2 streams at node i during time period h. The reader should refer to Fig. 1 where node and time period subscripts were dropped for clarity. For state change processes from state ξ to state ζ at node i, the ratio of outlet CO_2 flow over inlet CO_2 flow is denoted by parameter $\mu^{\zeta}_{\xi,i}$, where ξ= gas, liq, sup; ζ = gas, liq, sup, res. Here, "res" denotes the state of the CO_2 at well conditions for reservoir injection.

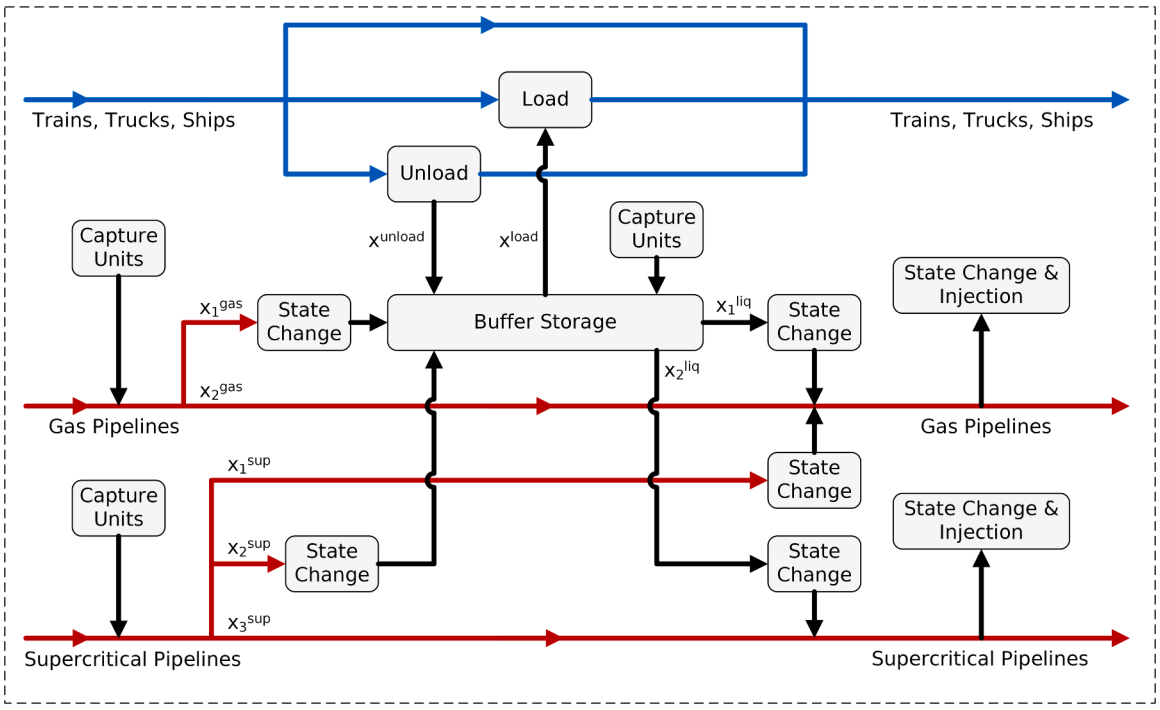


Fig. 1. Superstructure of processes and CO_2 flows modelled at emitters, inland geological reservoirs, ports and transshipment nodes. Dotted line: node boundary.

$$\sum_{(k,j,c):(k,j,i,c)\in N_{gas}} x^{p}_{k,c,h} + 1_{i\in I_e} \sum_{m:(i,m)\in I^{gas}_{M}} a_{i,m,h} = x^{gas}_{1,i,h} + x^{gas}_{2,i,h}, \forall i \in I\setminus\left(I^{sea}_{g}\cup I_{mix}\right), h\in H \tag{3}$$

$$x^{gas}_{2,i,h} + \mu^{gas}_{liq,i} x^{liq}_{1,i,h} + \mu^{gas}_{sup,i} x^{sup}_{1,i,h} = \sum_{(k,j,c):(k,i,j,c)\in N_{gas}} x^{p}_{k,c,h} + \frac{1_{i\in I^{inl}_{g}}}{\mu^{res}_{gas,i}} b^{gas}_{i,h}, \tag{4}$$

$$\forall i \in I\setminus\left(I^{sea}_{g}\cup I_{mix}\right), h\in H$$

$$\sum_{(k,j,c):(k,j,i,c)\in N_{sup}} x^{p}_{k,c,h} + 1_{i\in I_e} \sum_{m:(i,m)\in I^{sup}_{M}} a_{i,m,h} = x^{sup}_{1,i,h} + x^{sup}_{2,i,h} + x^{sup}_{3,i,h}, \tag{5}$$

$$\forall i \in I\setminus\left(I^{sea}_{g}\cup I_{mix}\right), h\in H$$

$$x^{sup}_{3,i,h} + \mu^{sup}_{liq,i} x^{liq}_{2,i,h} = \sum_{(k,j,c):(k,i,j,c)\in N_{sup}} x^{p}_{k,c,h} + \frac{1_{i\in I^{inl}_{g}}}{\mu^{res}_{sup,i}} b^{sup}_{i,h}, \forall i \in I\setminus\left(I^{sea}_{g}\cup I_{mix}\right), h\in H \tag{6}$$

$$\sum_{(k,j,c):(k,i,j,c)\in N_{\xi}} x^{p}_{k,c,h} - \sum_{(k,j,c):(k,j,i,c)\in N_{\xi}} x^{p}_{k,c,h} = 0, \forall i \in I_{mix}, h\in H, \xi \in \{gas, sup\} \tag{7}$$

$$f_{i,h} = f_{i,h-1} + \mu^{liq}_{gas,i} x^{gas}_{1,i,h} + \mu^{liq}_{sup,i} x^{sup}_{2,i,h} + \sum_{v\in V:(i,v)\in U} x^{\text{unload}}_{i,v,h} - \sum_{v\in V:(i,v)\in L} x^{\text{load}}_{i,v,h}$$

$$- x^{liq}_{1,i,h} - x^{liq}_{2,i,h} + 1_{i\in I_e} \sum_{m\in M:(i,m)\in I^{liq}_{M}} a_{i,m,h}, \tag{8}$$

$$\forall i \in I\setminus\left(I^{sea}_{g}\cup I_{mix}\right), h\in H$$

The capacity of a state change process is defined as its maximum annual *inlet* flow. This inlet flow corresponds to the previously defined variable $x^{\xi}_{\varsigma,i,h}$ (Fig. 1), where $\zeta = 1$ if ξ = gas; $\zeta = 1$ or 2 if ξ = sup or liq. The corresponding state change process capacity can be increased incrementally at every time period h via the variable $q^{\xi}_{\varsigma,i,h}$. Eq. 9 ensures that before a site is selected, no state change capacity can be added. Eq. 10 bounds the inlet annual flow ($x^{\xi}_{\varsigma,i,h}$) to a state change process at a given time period by the cumulative capacity added in previous time periods (including the current one). For inland nodes without inlet and outlet gas pipeline arcs, the variables $q^{sup}_{1,i,h}$ and $q^{liq}_{1,i,h}$can be fixed at zero since no gas phase transition is considered (e.g. by setting their upper bounds to zero). The parameter Q^{ξ}_{i} represents an upper limit (big-M constant) on the additional state change capacity $q^{\xi}_{\varsigma,i,h}$.

$$q^{\xi}_{\varsigma,i,h} \le Q^{\xi}_{i} \sum_{\tau=1}^{h} y_{i,\tau}, \forall i \in I\setminus\left(I^{sea}_{g}\cup I_{mix}\right), h\in H,\ \xi \in \{gas, sup, liq\}, \tag{9}$$

$$\varsigma = 1,\ if\ \xi = gas;\ \varsigma = 1\ or\ 2,\ if\ \xi = sup\ or\ liq$$

$$x^{\xi}_{\varsigma,i,h} \le \sum_{\tau=1}^{h} q^{\xi}_{1,i,\tau}, \forall i \in I\setminus\left(I^{sea}_{g}\cup I_{mix}\right), h\in H,\ \xi \in \{gas, sup, liq\}, \tag{10}$$

$$\varsigma = 1,\ if\ \xi = gas;\ \varsigma = 1\ or\ 2,\ if\ \xi = sup\ or\ liq.$$

4.3.2. Capture

Each emitter site can include multiple capture units. Over the whole time horizon, each capture unit can only be introduced once (Eq. 13) and cannot have its annual capture rate exceed its capacity (Eq. 12). In particular, if a capture unit was not introduced in a previous time period, then the sum on the right-hand-side of Eq. 12 vanishes and the corresponding annual capture rate is forced to zero. The maximal capturable CO_2 emissions for capture unit m at emitter site i during period h are represented by parameter $Q^{e}_{i,m,h}$. To reflect the fact that emission profiles in some sectors may change over time, due to industrial process changes or plants closing, the parameter $Q^{e}_{i,m,h}$ can vary with respect to time periods. The annual capture rates of all capture units at an emitter site are aggregated in Eq. 11 (variable $\hat{a}_{i,h}$). Binary variables denoting the introduction of a capture unit ($y^{e}_{i,m,h}$) and site selection ($y_{i,h}$) are linked in Eq. 14, where |.| stands for the cardinality of a set. This equation states that the number of capture units introduced at a time period cannot exceed the total number of potential capture units for that site. The total

annual capture rate of all capture units at site i during period h is denoted by variable $\widehat{a}_{i,h}$.

$$\widehat{a}_{i,h} = \sum_{m \in M:(i,m) \in I_M} a_{i,m,h}, \forall i \in I_e, h \in H \tag{11}$$

$$a_{i,m,h} \leq Q^e_{i,m,h} \sum_{\tau=1}^{h} y^e_{i,m,\tau}, \forall (i,m) \in I_M, h \in H \tag{12}$$

$$\sum_{h \in H} y^e_{i,m,h} \leq 1, \forall (i,m) \in I_M \tag{13}$$

$$\sum_{m \in M:(i,m) \in I_M} y^e_{i,m,h} \leq |\{m \in M : (i,m) \in I_M\}| \sum_{\tau=1}^{h} y_{i,\tau}, \forall i \in I_e, h \in H \tag{14}$$

4.3.3. Geological storage

At a reservoir site (inland or offshore), the annual injection rate during a time period is decomposed (Eq. 15) as the sum of the annual injection rate coming from pipelines and from ships (for offshore sites). The annual injection rate coming from pipelines is in turn partitioned to distinguish its origin, from gas or supercritical pipelines (Eq. 16), since conditioning processes (and costs) are different depending on CO_2 state. At both inland and offshore reservoirs, during a time period, total annual injection rates are bounded by injectivity (Eq. 17). The injectivity parameter $Q^g_{i,h}$ can vary with respect to time periods to reflect its decrease over time due to precipitation near the injection point, pressure effects or other factors causing injection rates to decay (Lane et al., 2021). For each reservoir, the total quantity of injected CO_2 over the time horizon can't exceed the site's operational lifetime capacity (Eq. 18). Stacked reservoirs can also be modelled by decomposing a node into multiple sub-nodes (see Appendix B). The variable $\widehat{b}_{i,h}$ denotes the annual injection rate at reservoir i during period h cumulative over pipelines and ships. The injection rates coming from gas and supercritical pipelines are denoted by $b^{gas}_{i,h}$ and $b^{sup}_{i,h}$, respectively (Fig. 1). The duration of time period h is represented via parameter O_h. The reservoir injectivity at time period h is indicated by parameter $Q^g_{i,h}$ and the total reservoir capacity is expressed via parameter Q^{res}_i.

$$\widehat{b}_{i,h} = b^p_{i,h} + 1_{i \in I^{sea}_g} b^s_{i,h}, \forall i \in I_g, h \in H \tag{15}$$

$$b^p_{i,h} = b^{gas}_{i,h} + b^{sup}_{i,h}, \forall i \in I_g, h \in H \tag{16}$$

$$\widehat{b}_{i,h} \leq Q^g_{i,h} \sum_{\tau=1}^{h} y_{i,\tau}, \forall i \in I_g, h \in H \tag{17}$$

$$\sum_{h \in H} O_h \widehat{b}_{i,h} \leq Q^{\text{res}}_i \sum_{h \in H} y_{i,h}, \forall i \in I_g \tag{18}$$

4.3.4. Offshore geological storage

Transporting CO_2 to offshore reservoirs can be done via ships or pipelines laid on the seabed. Ships can either be connected directly to the injection points (direct injection) or can unload to intermediate platforms (FSI), which temporarily store CO_2 and are linked to the injection point (indirect injection). For direct injection, ships need to connect and disconnect to pipe risers, which are attached to the injection points (Nakazawa et al., 2013). This leads to a batchwise injection process. In some cases, ship queues may form for pipe riser access (Choi and Chang, 2011). Moreover, it is anticipated that well injectivity will be significantly inferior to normal ship cargo pump unloading rates (CCSa and ZEP, 2023). Therefore, in the case of direct injection, injectivity may be an important limiting factor to the number of ship trips and CO_2 volumes transported. This issue is alleviated by the introduction of buffer storage. In particular, indirect injection via an FSI can be done continuously, assuming sufficient shipping volumes and FSI capacity. Not being limited by well injectivity, unloading to an FSI can be done at a higher rate (IEAGHG, 2020). This may lead to larger annual transported volumes and higher injection rates for indirect injection, at the expense of larger capital investment for the FSI. The frequency of harsh and benign water conditions also impacts the choice of unloading technology (IEAGHG, 2020). Indirect injection can maintain injection even when ships are unavailable due to inclement sea conditions, while direct injection operations must be interrupted. Direct and indirect injection involve conditioning the liquid CO_2 being transported before it can be injected in supercritical phase, i.e. raising its temperature and pressure to wellhead specifications to avoid the creation of reservoir fractures. Ships used for direct injection, termed henceforth "injection ships", have on-board conditioning, as well as anchoring and positioning systems. On the other hand, indirect injection involves "transport ships" that offload liquid CO_2 to an FSI, where conditioning takes place. The proposed optimization model permits injection by offshore pipelines and by ships (direct or indirect).

Fig. 2 summarizes the CO_2 flows at an offshore storage site. As in Fig. 1, for visual clarity some arrows were aggregated and some variable indices were dropped. When direct injection is selected, the FSI block in Fig. 2 vanishes and the "state change" block represents the conditioning occurring on ships before injection. On the other hand, when indirect injection is considered, ships unload to an FSI and the state change block then corresponds to the conditioning occurring on the FSI.

Mass balances for ship unloading and pipelines (gas, supercritical) correspond to Eqs. 19 to 21, respectively. Eq. 22 defines annual FSI buffer storage levels. Recall that the superscript *ind* and *dir* refer to indirect and direct injection, respectively.

$$\sum_{(k,j):(k,j,i,v) \in N_V} x_{k,v,h} = \sum_{(k,j):(k,i,j,v) \in N_V} x_{k,v,h}$$

$$+1_{(i,v) \in U} x^{\text{unload}}_{i,v,h}, \forall (i,v) \in I_V, h \in H \text{ with } i \in I^{sea}_g \tag{19}$$

$$\sum_{(k,j,c):(k,j,i,c) \in N_{gas}} x^p_{k,c,h} = x^{gas}_{1,i,h} + \sum_{(k,j,c):(k,i,j,c) \in N_{gas}} x^p_{k,c,h}, \forall i \in I^{sea}_g, h \in H \tag{20}$$

$$\sum_{(k,j,c):(k,j,i,c) \in N_{sup}} x^p_{k,c,h} = x^{sup}_{1,i,h} + \sum_{(k,j,c):(k,i,j,c) \in N_{sup}} x^p_{k,c,h}, \forall i \in I^{sea}_g, h \in H \tag{21}$$

$$f_{i,h} = f_{i,h-1} + \sum_{v \in V:(i,v) \in U} x^{\text{unload}}_{i,v,h} - \frac{1}{\mu^{dir}_{liq,i}} b^{dir}_{i,h} - \frac{1}{\mu^{ind}_{liq,i}} b^{ind}_{i,h}, \forall i \in I^{sea}_g, h \in H \tag{22}$$

At an offshore reservoir (Fig. 2), the annual injection rate coming from gas and supercritical pipelines (respectively) is assumed to be a proportion of the inlet flow to state change processes (μ^{res}_{gas} and μ^{res}_{sup}), such as (re)conditioning (Eqs. 23 and 24, respectively). Injection points linked to ships can either correspond to direct or indirect (FSI) injection (Eq. 25), but not simultaneously. To allow both ship injection modes concurrently, a junction node can be used to split a reservoir into two reservoir nodes, one for direct injection and the other indirect injection.

For each time period at each offshore storage site, the total annual injection rate coming from ships is split into direct and indirect terms (Eq. 25), each having its maximal annual injection rate limited by reservoir injectivity (Eqs. 26 and 27, respectively). The injection rate coming from ships ($b^s_{i,h}$) is split into two contributions coming from direct injection ($b^{dir}_{i,h}$) and indirect injection ($b^{ind}_{i,h}$).

$$b^{gas}_{i,h} = \mu^{res}_{gas,i} x^{gas}_{1,i,h}, \forall i \in I^{sea}_g, h \in H \tag{23}$$

$$b^{sup}_{i,h} = \mu^{res}_{sup,i} x^{sup}_{1,i,h}, \forall i \in I^{sea}_g, h \in H \tag{24}$$

$$b^s_{i,h} = b^{dir}_{i,h} + b^{ind}_{i,h}, \forall i \in I^{sea}_g, h \in H \tag{25}$$

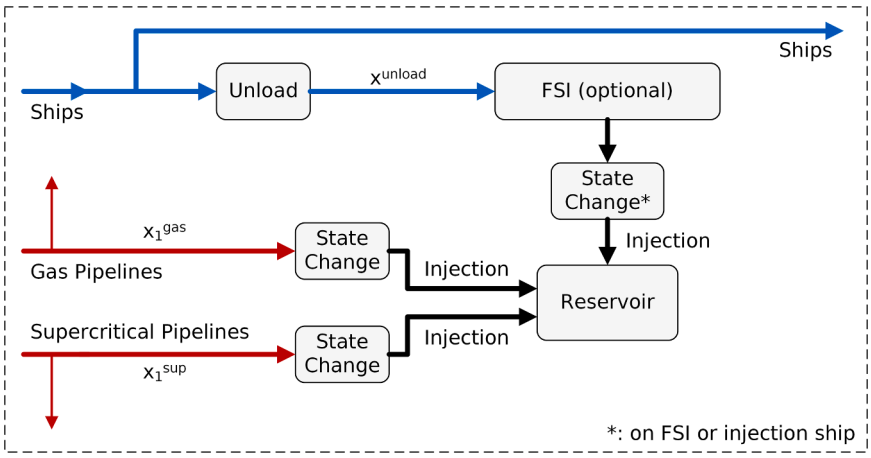


Fig. 2. Superstructure of processes and CO_2 flows modelled inside each offshore reservoir node. Dotted line: node boundary.

$$b^{dir}_{i,h} \leq Q^{g}_{i,h} \sum_{\tau=1}^{h} y^{dir}_{i,\tau}, \forall i \in I^{sea}_{g}, h \in H \tag{26}$$

$$b^{ind}_{i,h} \leq Q^{g}_{i,h} \sum_{\tau=1}^{h} y^{ind}_{i,\tau}, \forall i \in I^{sea}_{g}, h \in H \tag{27}$$

We assume that over the whole time horizon, direct or indirect injection can be introduced at most once at an offshore reservoir, but not both (Eq. 28). The chosen injection mode becomes available henceforth. The binary variables related to direct and indirect injection become zero if a site was not opened previously (Eqs. 29 and 30). If indirect injection was not previously selected, then transport ships cannot unload at an offshore storage site (Eq. 31). The parameter $Q^{unload}_{i,v,h}$ represents the upper limit on the annual unloading rate of the whole fleet of vehicles of type v at node i during period h (big-M parameter).

$$\sum_{h \in H} \left(y^{dir}_{i,h} + y^{ind}_{i,h} \right) \leq 1, \forall i \in I^{sea}_{g} \tag{28}$$

$$y^{dir}_{i,h} \leq \sum_{\tau=1}^{h} y_{i,\tau}, \forall i \in I^{sea}_{g}, h \in H \tag{29}$$

$$y^{ind}_{i,h} \leq \sum_{\tau=1}^{h} y_{i,\tau}, \forall i \in I^{sea}_{g}, h \in H \tag{30}$$

$$x^{\text{unload}}_{i,v,h} \leq Q^{unload}_{i,v,h} \sum_{\tau=1}^{h} y^{ind}_{i,\tau}, \forall (i,v) \in U, h \in H \text{ with } i \in I^{sea}_{g}, v \in V^{ind}_{s} \tag{31}$$

In order to assess ship fleet size and fleet mix, the period of activity of all ships of each type must be determined. In particular, the duration of unloading operations at offshore storage sites must be assessed. Therefore, we introduce the variable $\theta_{i,v,h}$ to represent the annual fleet unloading time at a reservoir *i* during period *h*, cumulative over all ships of type *v*. In the equations below, the parameters $\widetilde{Q}^{dir}_{i,v,h}$ and $\widetilde{Q}^{ind}_{i,v}$ represent the unloading rate of a *single* ship of type v for direct and indirect injection at reservoir i, respectively. The time period index h on $\widetilde{Q}^{dir}_{i,v,h}$ is included to accommodate for injection limit forecasts that could constrain ship unloading rates for direct injection. For transport ships, the variable $\theta_{i,v,h}$ is defined via the following Equation.

$$\theta_{i,v,h} = \frac{1}{\widetilde{Q}^{ind}_{i,v}} x^{\text{unload}}_{i,v,h}, \forall (i,v) \in U, h \in H \text{ with } i \in I^{sea}_{g}, v \in V^{ind}_{s} \tag{32}$$

For injection ships, the cumulative unloading time at an offshore reservoir is given by the following nonlinear Eq. (33). In terms of units of measurement, the right-hand-side of this Eq. is multiplied by 1 year, i.e. we're considering the total CO_2 tonnage unloaded in a year.

$$\theta_{i,v,h} = \frac{1}{\widetilde{Q}^{dir}_{i,v,h}} x^{\text{unload}}_{i,v} + \left(\frac{1}{\widetilde{Q}^{ind}_{i,v}} - \frac{1}{\widetilde{Q}^{dir}_{i,v,h}} \right) x^{\text{unload}}_{i,v,h} \sum_{\tau=1}^{h} y^{ind}_{i,\tau}, \ \forall (i,v) \in U \text{ with } i \in I^{sea}_{g} \text{ and } v \in V^{dir}_{s} \tag{33}$$

Note that the ship unloading rate for direct injection depends on h since reservoir injectivities may change over time. This nonlinear non-convex equation involves the product of a continuous variable ($x^{unload}_{i,v,h}$) and binary variables ($y^{ind}_{i,\tau}$). In order to remain in the linear (MILP) realm, the following reformulation is proposed (Eqs. 34 to 37). We introduce the variable $x^{ind}_{i,v,h}$, which is a linear reformulation of the product $x^{unload}_{i,v,h} \sum_{\tau=1}^{h} y^{ind}_{i,\tau}$. To see this, note that Eq. 28 forces the sum $\sum_{\tau=1}^{h} y^{ind}_{i,\tau}$ to only take values 0 or 1. It follows by substituting those values in Eqs. 34 to 36 that $x^{ind}_{i,v,h}$ is equal to $x^{unload}_{i,v,h}$ when $\sum_{\tau=1}^{h} y^{ind}_{i,\tau}$ takes value 1 and is zero otherwise. Eq. 33 can thus be rewritten linearly using this new variable (Eq. 37).

$$x^{ind}_{i,v,h} \leq Q^{unload}_{i,v,h} \sum_{\tau=1}^{h} y^{ind}_{i,\tau}, \forall (i,v) \in U, h \in H \text{ with } i \in I^{sea}_{g}, v \in V^{dir}_{s} \tag{34}$$

$$x^{ind}_{i,v,h} \leq x^{\text{unload}}_{i,v,h}, \forall (i,v) \in U, h \in H \text{ with } i \in I^{sea}_{g}, v \in V^{dir}_{s} \tag{35}$$

$$x^{ind}_{i,v,h} \geq x^{\text{unload}}_{i,v,h} - Q^{unload}_{i,v,h} \left(1 - \sum_{\tau=1}^{h} y^{ind}_{i,\tau} \right), \forall (i,v) \in U, h \in H \text{ with } i \in I^{sea}_{g}, v \in V^{dir}_{s} \tag{36}$$

$$\theta_{i,v,h} = \frac{1}{\widetilde{Q}^{dir}_{i,v,h}} x^{\text{unload}}_{i,v,h} + \left(\frac{1}{\widetilde{Q}^{ind}_{i,v}} - \frac{1}{\widetilde{Q}^{dir}_{i,v,h}} \right) x^{ind}_{i,v,h}, \forall (i,v) \in U, h \in H \text{ with } i \in I^{sea}_{g}, v \in V^{dir}_{s} \tag{37}$$

4.3.5. *Buffer storage and FSI sizing*

Buffer storage tanks are an integral part of CCUS value chains with multimodal transport, linking continuous processes (e.g. capture, pipeline transportation) to batchwise vehicle transportation. Depending on the layout of the space available, spherical or cylindrical tanks (vertical or horizontal) can be used (Fraga et al., 2021). Besides process engineering considerations (e.g. insulation, pumping), hourly vehicle

schedules play an integral part in determining buffer storage capacity (Lee et al., 2012). Since strategic planning considers timescales that are several orders of magnitude higher than vehicle scheduling, detailed buffer storage sizing based on vehicle scheduling falls outside the purview of the present study.

Sizing of storage tanks must also account for the impact of potential disruptions, such as interruptions in processes operations (capture, storage) or vehicles due to unplanned maintenance. As a consequence, an excess capacity safety margin is added as a precautionary measure. For shipping, multiple studies recommend total buffer storage tank capacity between 1 and 2 times the maximal load of a single ship (Seo et al., 2017), while others add a safety margin (e.g. 120 %) based on the total fleet capacity (Yoo, 2013; Element Energy, 2018). In a report by the IEAGHG (2023), the authors consider various value chain examples to size storage tank capacity. As a rule of thumb, they define buffer storage capacity enough to store 2 days of operations.

In the context of general CCUS value chains, we propose the following equations to determine buffer storage capacity and annual levels. Annual levels and available capacity at the beginning of the planning horizon are dictated by Eqs. 38 and 39, respectively. These equations, which also apply to FSI, could be particularly useful if the model is used in a rolling horizon framework. Capacity for buffer storage and FSI can be added at each time period within predefined limits (Eqs. 41 and 42, respectively). The annual level at a time period is limited by the installed capacity until that time period (Eq. 40). The buffer storage or FSI capacity that can be added at the beginning of period h at site i is represented by variable $\widehat{f}_{i,h}$, with lower and upper limit parameters $\widehat{f}_i^{min}$ and $\widehat{f}_i^{max}$, respectively.

$$f_{i,0} = F_i^0, \forall i \in I_e \cup I_g \cup I_{hub} \cup I_s \tag{38}$$

$$\widehat{f}_{i,0} = \widehat{F}_{i,0}, \forall i \in I_e \cup I_g \cup I_{hub} \cup I_s \tag{39}$$

$$f_{i,h} \leq \sum_{\tau=0}^{h} \widehat{f}_{i,\tau}, \forall i \in I_e \cup I_g \cup I_{hub} \cup I_s, h \in H \tag{40}$$

$$\widehat{f}_i^{min} \sum_{\tau=1}^{h} y_{i,\tau} \leq \widehat{f}_{i,h} \leq \widehat{f}_i^{max} \sum_{\tau=1}^{h} y_{i,\tau}, \forall i \in I_e \cup I_g^{inl} \cup I_{hub} \cup I_s, h \in H \tag{41}$$

$$\widehat{f}_i^{min} \sum_{\tau=1}^{h} y_{i,\tau}^{ind} \leq \widehat{f}_{i,h} \leq \widehat{f}_i^{max} \sum_{\tau=1}^{h} y_{i,\tau}^{ind}, \forall i \in I_g^{sea}, h \in H \tag{42}$$

We generalize the buffer storage sizing rule of thumb from IEAGHG (2023). More precisely, for a given inland site *i* and a given time period *h*, we assume that the total buffer storage capacity must be at least the maximum between inlet and outlet flow over a given duration (e.g. 2 days) with a safety margin factor (e.g. 1.2). The product of this duration and safety margin is encapsulated in the parameter β_i. Eqs. 43 and 44 summarize this rule of thumb.

$$\sum_{\tau=0}^{h} \widehat{f}_{i,\tau} \geq \beta_i \left(\mu_{gas,i}^{liq} x_{1,i,h}^{gas} + \mu_{sup,i}^{liq} x_{2,i,h}^{sup} + \sum_{v \in V:(i,v) \in U} x_{i,v,h}^{\text{unload}} + 1_{i \in I_e} \sum_{m \in M:(i,m) \in I_M^{liq}} a_{i,m,h} \right), \tag{43}$$

$$\forall i \in I \backslash \left(I_g^{sea} \cup I_{mix} \right), h \in H$$

$$\sum_{\tau=0}^{h} \widehat{f}_{i,\tau} \geq \beta_i \left(\sum_{v \in V:(i,v) \in L} x_{i,v,h}^{\text{load}} + x_{1,i,h}^{liq} + x_{2,i,h}^{liq} \right), \forall i \in I \backslash \left(I_g^{sea} \cup I_{mix} \right), h \in H \tag{44}$$

The capacity of FSI is bounded in a similar manner (Eqs. 45 and 46).

$$\sum_{\tau=0}^{h} \widehat{f}_{i,\tau} \geq \beta_i \sum_{v \in V:(i,v) \in U} x_{i,v,h}^{ind}, \forall i \in I_g^{sea}, h \in H \tag{45}$$

$$\sum_{\tau=0}^{h} \widehat{f}_{i,\tau} \geq \frac{\beta_i}{\mu_{liq,i}^{ind}} b_{i,h}^{ind}, \forall i \in I_g^{sea}, h \in H \tag{46}$$

4.3.6. Pipeline capacities and flows

The cost of a pipeline is inherently a function of its diameter and length. However, modelling pipeline diameters explicitly in a MILP framework can lead to prohibitive calculation times and poor results (Middleton, 2013). To address this issue, we employ the piecewise linearization of pipeline costs from Middleton (2013), where diameters are replaced with capacities, i.e. maximal flows (Mtonnes/year). The capacity upper limit of each line segment is defined in Eq. 47. On a given pipeline arc at a given time period, the pipeline flow is limited by the total installed capacity thus far (Eq. 48).

$$q_{k,c,h} \leq Q_{k,c}^{max} y_{k,c,h}^{p}, \forall (k,c) \in K_C, h \in H \tag{47}$$

$$x_{k,c,h}^{p} \leq \sum_{\tau=1}^{h} q_{k,c,\tau}, \forall (k,c) \in K_C, h \in H \tag{48}$$

4.3.7. Node selection constraints

A site can be opened at most once during the time horizon (Eq. 49). If it was not opened up to a given time period, then the vehicle loading and unloading rates vanish (Eqs. 50 and 51). These equations, coupled with Eqs. 9 and 10 imply that the annual internal flows (Fig. 1) of gas, supercritical and liquid CO_2 vanish at a site if it was not previously opened. The loading and unloading rates at node i for the fleet of vehicles of type v during period h are bounded above by parameters $Q_{i,v}^{load}$ and $Q_{i,v,h}^{unload}$ respectively (big-M constants).

$$\sum_{h \in H} y_{i,h} \leq 1, \forall i \in I_e \cup I_g \cup I_{hub} \cup I_s \tag{49}$$

$$x_{i,v,h}^{\text{load}} \leq Q_{i,v}^{load} \sum_{\tau=1}^{h} y_{i,\tau}, \forall (i,v) \in L \text{ with } i \in I_e \cup I_g^{inl} \cup I_{hub} \cup I_s, h \in H \tag{50}$$

$$x_{i,v,h}^{\text{unload}} \leq Q_{i,v,h}^{unload} \sum_{\tau=1}^{h} y_{i,\tau}, \forall (i,v) \in U \text{ with } i \in I_e \cup I_g \cup I_{hub} \cup I_s, h \in H \tag{51}$$

4.3.8. Vehicles: flows, loads and trips

The fleet sizing and fleet mix model proposed in this study follows closely from those of Fraga et al. (2021) and Nie et al. (2021), which were used to carry out preliminary assessments for the Grenland hub and Northern Lights project in Norway. We make the standard assumption that over the span of a year, vehicles make round trips, that is, they eventually return to their point of origin by the end of the year. Not doing so would require the scheduling and routing of each

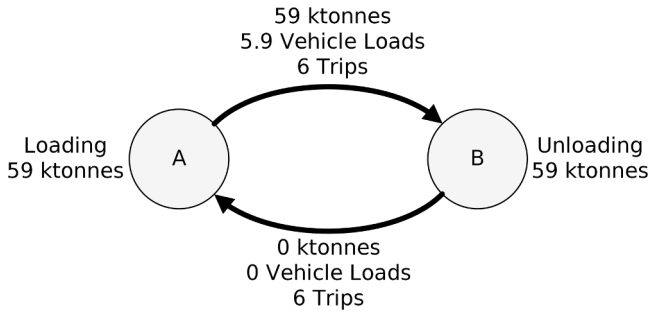


Fig. 3. Example illustrating vehicle fleet loads and trips (single vehicle capacity: 10 ktonnes).

individual vehicle (as opposed to fleets), which falls outside the scope of strategic planning and falls under operational considerations.

To illustrate the concept of vehicle fleet loads and trips, consider the example depicted in Fig. 3. To simplify, we assume a single period of a year. A fleet of vehicles each having a capacity of 10 ktonnes picks up 59 ktonnes of captured CO_2 at emitter A, transports it to reservoir B, where it unloads 59 ktonnes, before returning back to the emitter. In that context, we say that this fleet transports 5.9 vehicle loads from A to B and 0 from B to A (vehicles initially assumed empty). This corresponds to a total of 6 vehicle trips going from A to B and since we assume round trips, the same is true from B to A. Note that this does not imply that exactly 6 vehicles are needed, since the actual number of vehicles required may be lower, depending on their speed, (un)loading rates and other factors (see next subsection).

Contrary to this example, the proposed model can cover any topology for round trips (i.e. any graph cycle) as well as partial (un)loading at nodes. During a time period, the annual flow of the fleet of vehicles of a given type on a given arc is denoted by $x_{k,v,h}$. This corresponds to a number of vehicle loads ($l_{k,v,h}$), defined by dividing the flow by the cargo capacity of a single vehicle (Q_v^{cap}) – see Eq. 52. The number of trips (or "outings") is defined via integer variable $n^o_{k,v,h}$ and Eq. 53. Since we assume round trips, the number of inlet and outlet trips at a node must be equal (Eq. 54).

$$x_{k,v,h} = Q_v^{\text{cap}} l_{k,v,h}, \forall (k,v) \in K_V, h \in H \tag{52}$$

$$n^o_{k,v,h} \geq l_{k,v,h}, \forall (k,v) \in K_V, h \in H \tag{53}$$

$$\sum_{(k,j):(k,i,j,v)\in N_V} n^o_{k,v,h} = \sum_{(k,j):(k,j,i,v)\in N_V} n^o_{k,v,h}, \forall (i,v) \in I_V, h \in H \tag{54}$$

4.3.9. Fleet sizing and fleet mix

In order to approximate the annual number of vehicles of a given type needed during a time period, fleet sizing and fleet mix models (Fraga et al., 2021) determine the annual cumulative duration of activity (in years) of the fleet during that time period ($\widehat{\theta}_{v,h}$). For rail and road transportation, the period of fleet activity in a year is split between travelling, loading and unloading. Off season and general down time (e.g. due to weather conditions) falls into the category of inactive time (Fig. 4a). For ships, active time also includes being berthed at ports and offshore reservoirs (Fig. 4b). For inland and offshore vehicle transportation respectively, Eqs. 55 and 56 define the annual period of activity, summed over all vehicles of a given type during each time period (Fig. 4).

The annual number of vehicles of a given type needed to transport CO_2 during a time period ($n_{v,h}$) is obtained by dividing the annual cumulative activity period of the corresponding fleet ($\widehat{\theta}_{v,h}$) by the maximal proportion of a year a vehicle of this type can be active, rounded up (Eq. 57). In the proposed model, a distinction is made between vehicles already available at the beginning of the time horizon that don't need to be purchased (Eq. 58) and new vehicles that are purchased in each time period (Eq. 59). Terms on the right-hand side of Eqs. 55 and 56 correspond to the vehicle activity types listed in Fig. 4. The length of arc k (shipping lane, railroad or road) is denoted by parameter δ_k while the average speed of the fleet of vehicles of type v on arc k is represented by parameter $\sigma_{k,v}$. The proportion of a year where vehicles can be active is symbolized as τ_v^{max} (parameter in the interval [0,1]).

$$\widehat{\theta}_{v,h} = \sum_{k\in K:(k,v)\in K_V} \frac{\delta_k n^o_{k,v,h}}{\sigma_{k,v}} + \sum_{i\in I:(i,v)\in L} \frac{x^{\text{load}}_{i,v,h}}{\widetilde{Q}^{load}_{i,v}} + \sum_{i\in I:(i,v)\in U} \frac{x^{\text{unload}}_{i,v,h}}{\widetilde{Q}^{unload}_{i,v,h}}, \tag{55}$$

$\forall v \in V_r \cup V_t, h \in H$

$$\widehat{\theta}_{v,h} = \sum_{k\in K:(k,v)\in K_V} \frac{\delta_k n^o_{k,v,h}}{\sigma_{k,v}} + \sum_{i\in I:(i,v)\in L} \frac{x^{\text{load}}_{i,v,h}}{\widetilde{Q}^{load}_{i,v}} + \sum_{i\in I\setminus I^{sea}_g:(i,v)\in U} \frac{x^{\text{unload}}_{i,v,h}}{\widetilde{Q}^{unload}_{i,v,h}} \tag{56}$$

$$+ \sum_{i\in I^{sea}_g:(i,v)\in U} \theta_{i,v,h} + \sum_{i\in I:(i,v)\in I_V} \tau^{berth}_{i,v} \sum_{(k,j):(k,i,j,v)\in N_V} n^o_{k,v,h}, \forall v \in V_s, h \in H$$

$$\widehat{\theta}_{v,h} \leq \tau_v^{max} n_{v,h}, \forall v \in V, h \in H \tag{57}$$

$$n^{\text{new}}_{v,0} = N^0_v, \forall v \in V \tag{58}$$

$$n_{v,h} \leq \sum_{\tau=0}^{h} n^{\text{new}}_{v,\tau}, \forall v \in V, h \in H \tag{59}$$

4.3.10. Rail and road extensions

Railways and roads may need to be extended in order to reach specific emitters, reservoirs, ports or transshipment hubs. Network arcs corresponding to potential rail or road extensions are grouped in the set K^{rt}. If such an extension is not built, then the vehicle flows on that arc vanish (Eq. 60). Each extension can only be constructed once during the time horizon (Eq. 61). The choice to add a road or railroad extension corresponding to arc k in K^{rt} at the beginning of period h is modelled via binary variable $y^{rt}_{k,h}$, which takes value 1 if and only if the extension is added.

$$\sum_{v:(k,v)\in K_V} x_{k,v,h} \leq Q^{rt}_k \sum_{\tau=1}^{h} y^{rt}_{k,\tau}, \forall k \in K^{rt}, h \in H \tag{60}$$

$$\sum_{h\in H} y^{rt}_{k,h} \leq 1, \forall k \in K^{rt} \tag{61}$$

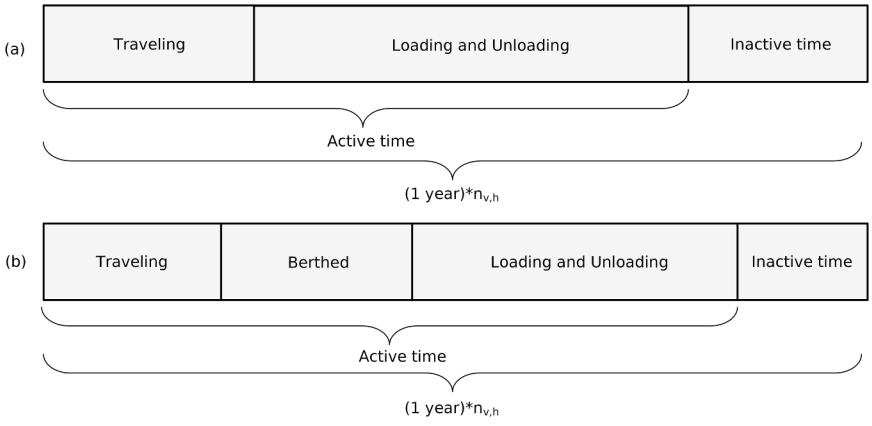


Fig. 4. Active and inactive time for trains and trucks (a) and ships (b).

4.3.11. Capture and sequestration targets

Similarly to CO_2 pipeline models (Jones et al., 2022), annual capture targets ($CapCO2_h$) can be specified at each time period (Eq. 62). However, such models do not include buffer storage, therefore the total annual CO_2 injected equals the total annual CO_2 captured. Due to the presence of buffer storage in the proposed model, injection targets ($InjCO2_h$) must be explicitly enforced (Eq. 63).

$$\sum_{i\in I_e} \widehat{a}_{i,h} = \mathrm{CapCO2}_h, \forall h \in H \tag{62}$$

$$\sum_{i\in I_g} \widehat{b}_{i,h} = \mathrm{InjCO2}_h, \forall h \in H \tag{63}$$

4.3.12. Costs and objective function

The proposed model aims to minimize the total cost (TC) for capture, transportation and storage (Eq. 64). There are various ways to partition these costs. The decomposition proposed below aims to group costs by their type. However, in some cases (e.g. TC_{site}) costs are grouped according to the model variables they multiply in order to ensure a smaller number of terms in the objective function. We use the terms "fixed" and "variable" cost depending on if they involve integer/binary or continuous variables (respectively). Both fixed and variable costs can incorporate a mix of CAPEX and OPEX and are adjusted in the multiperiod context using discount rates and amortization similarly to Middleton et al. (2012). Carbon taxes on emissions can be considered by adjusting the cost parameters of capture units accordingly, as suggested in Jones et al. (2022). The remainder of this section explains each term of Eq. 64.

$$TC = TC_p + TC_{rt} + TC_V + TC_e + TC_g + TC_{site} + VC_{handling} + VC_{state} \tag{64}$$

Pipeline costs

The total pipeline cost (TC_P) is given by a piecewise linear approximation (Eq. 65) as in Middleton (2013). The parameters $VC^p_{k,c,h}$ and $FC^p_{k,c,h}$ correspond to variable and fixed pipeline costs (respectively) for arc k during period h.

$$TC_p = \sum_{(k,c)\in K_C, h\in H} \left(VC^p_{k,c,h} q_{k,c,h} + FC^p_{k,c,h} y^p_{k,c,h} \right) \tag{65}$$

New railways and new roads costs

The cost for the construction of new roads and railroads for truck and train transportation (TC_{rt}) is defined in Eq. 66, where parameter $FC^{rt}_{k,h}$ is the fixed cost for building rail spur or road k during period h.

$$TC_{rt} = \sum_{k\in K^{rt}, h\in H} FC^{rt}_{k,h} y^{rt}_{k,h} \tag{66}$$

Vehicle costs

The total vehicle cost (TC_V) is given by Eq. 67. After describing the parameters used in this new formulation, we show how it generalizes the vehicle cost functions of Han and Lee (2011a) and Fraga et al. (2021).

$$TC_V = \sum_{(k,v)\in K_V, h\in H} \left(FC^o_{k,v,h} n^o_{k,v,h} + VC^{rst}_{k,v,h} x_{k,v,h} \right) + \sum_{v\in V, h\in H} \left(FC^{rst}_{v,h} n_{v,h} + FC^{\mathrm{new}}_{v,h} n^{\mathrm{new}}_{v,h} + VC^{time}_{v,h} \widehat{\theta}_{v,h} \right) \tag{67}$$

This new formulation gives the flexibility to define vehicle costs in alternative ways, using the following parameters.

- The parameter $FC^o_{k,v,h}$ can be used to account for costs (OPEX and CAPEX) proportional to the annual number of vehicle trips of a given type along a route. These include road tolls and harbour costs. If a vehicle's fuel economy (km/L) is known, then $FC^o_{k,v,h}$ can also be used to model fuel costs by multiplying fuel price (M$/L) by arc length (km) and dividing by fuel economy. Vehicle maintenance costs that are proportional to the distance travelled annually can also be added in this parameter (multiplying the number of trips by arc length).
- The parameter $VC^{rst}_{k,v,h}$ can be used to incorporate costs (OPEX and CAPEX) proportional to the annual CO_2 tonnage transported by the fleet of vehicles of a given type on a given route, e.g. the fuel cost approximation of Han and Lee (2011) - see explanation below.
- The parameter $FC^{rst}_{v,h}$ can be used to take into account costs (OPEX and CAPEX) proportional to the number of vehicles of each type, excluding fixed costs related to the purchase of new vehicles. If $FC^{rst}_{v,h}$ is non-zero, then it follows that at a global minimum of the objective function we have the following relation (Eq. 70), where $\lceil . \rceil$ is the ceiling operator.

$$n_{v,h} = \left\lceil \widehat{\theta}_{v,h} / \tau^{max}_v \right\rceil, \forall v \in V, h \in H \tag{68}$$

- The parameter $FC^{new}_{v,h}$ corresponds to fixed costs related to the purchase of new vehicles.
- The parameter $VC^{time}_{v,h}$ can be used to consider costs (OPEX and CAPEX) proportional to the annual cumulative period of activity (in years) of the fleet of vehicles of type v (e.g. labor costs).

For detailed examples of vehicle cost breakdowns, the reader can refer to Han and Lee (2011a) and Fraga et al. (2021). For ships, see Eq. 5 in Fraga et al. (2021). For ships, trucks and trains, see Eqs. 6 to 10 and 18 to 28 in Han and Lee (2011a). The vehicle fuel cost model in Han and Lee (2011) is equivalent to assigning the following values:

$$VC^{rst}_{k,v,h} = \frac{\delta_k \rho_{v,h}}{Q^{cap}_v \epsilon_v} \tag{69}$$

where ϵ_v and $\rho_{v,h}$ correspond respectively to the fuel economy of vehicle type v (km/L of fuel) and fuel price for vehicle type v (M$/L). Moreover, the vehicle labour costs of Han and Lee (2011) are equivalent to setting $VC^{time}_{v,h}$ to the driver wage (in M$/year). The vehicle maintenance costs proposed by the same authors are equivalent to setting $FC^o_{k,v,h}$ to the product of maintenance expenses (M$/km) and arc length (km/trip). The shipping cost proposed in this work generalizes that of Fraga et al. (2021) by defining loading and unloading times as explicit functions of their rates, by adding berth times and by considering direct and indirect ship injection and their related ship durations.

The variable costs for vehicle loading and unloading at inland and offshore sites are given in Eq. 70, where $VC^{\mathrm{load}}_{i,v,h}$ and $VC^{\mathrm{unload}}_{i,v,h}$ are the corresponding cost parameters.

$$VC_{handling} = \sum_{(i,v)\in L, h\in H} VC^{\mathrm{load}}_{i,v,h} x^{\mathrm{load}}_{i,v,h} + \sum_{(i,v)\in U, h\in H} VC^{\mathrm{unload}}_{i,v,h} x^{\mathrm{unload}}_{i,v,h} \tag{70}$$

Capture costs

The total cost associated to capture units at emitter sites is given by Eq. 71, where parameters for fixed and variable capture costs are denoted by $FC^e_{i,m,h}$ and $VC^e_{i,m,h}$, respectively.

$$TC_e = \sum_{(i,m)\in I_M, h\in H} \left(FC^e_{i,m,h} y^e_{i,m,h} + VC^e_{i,m,h} a_{i,m,h} \right) \tag{71}$$

Geological storage costs

The total cost for geological storage is given by Eq. 72. The parameter $VC^g_{i,h}$ is associated to variable pipeline injection costs, while parameters $VC^{dir}_{i,h}$ and $VC^{ind}_{i,h}$ correspond to variable costs for direct and indirect injection, respectively. The parameters $FC^{dir}_{i,h}$ and $FC^{ind}_{i,h}$ relate to fixed costs for direct and indirect injection, respectively. On the right-hand side of Eq. 72, the first term contains the cost for injection at both inland and offshore reservoirs. Although wells are not counted explicitly in the model, drilling costs at each reservoir node can be added to parameter $VC^g_{i,h}$, e.g. using the correlations of McCollum and Ogden

(2006) linking those costs to injection rates. The second term lists the variable and fixed costs for direct and indirect injection via ships (including FSI fixed costs $FC^{\text{tank}}_{i,h}$).

$$TC_g = \sum_{i\in I_g, h\in H} VC^{g}_{i,h} b^{p}_{i,h} \tag{72}$$

$$+ \sum_{i\in I^{sea}_g, h\in H} \left(VC^{dir}_{i,h} b^{dir}_{i,h} + VC^{ind}_{i,h} b^{ind}_{i,h} + FC^{dir}_{i,h} y^{dir}_{i,h} + \left(FC^{ind}_{i,h} + FC^{\text{tank}}_{i,h} \right) y^{ind}_{i,h} \right)$$

Site specific costs

Opening a site (i) at the beginning of a time period (h) involves various fixed costs (parameter $FC^{site}_{i,h}$) for vehicle loading and unloading, buffer storage and CO_2 state change processes (see Appendix C for full breakdown of $FC^{site}_{i,h}$). It is assumed that these costs are incurred when a site is initially opened. Disaggregating these fixed costs to allow them to occur at time periods after a site is opened would require new binary variables for each cost type (see Appendix C) and each time period, which would increase the combinatorial complexity of the problem and may hamper computational performance. Therefore, this modelling direction was not considered. At inland and offshore sites, fixed costs ($FC^{site}_{i,h}$) that multiply site selection binaries as well as variable buffer storage and FSI costs are defined in Eq. 73. The parameter $VC^{\text{tank}}_{i,h}$ is associated to variable costs for CO_2 storage tanks or FSI (offshore reservoirs) at node i during period h. The user should be wary as to not double count the variable costs for FSI in both the $VC^{\text{tank}}_{i,h}$ and $VC^{ind}_{i,h}$ parameters, the latter relates to variable costs to inject CO_2 from an FSI into a reservoir while the former represents FSI variable costs (excluding injection costs).

$$TC_{site} = \sum_{i\in I_e\cup I_s\cup I_{hub}\cup I_g, h\in H} \left(FC^{site}_{i,h} y_{i,h} + VC^{\text{tank}}_{i,h} \widehat{f}_{i,h} \right) \tag{73}$$

State change costs

The variable cost for state change processes related to inland and offshore sites is given by Eq. 74. This includes conditioning occurring on injection ships and FSI (second term of Eq. 74). The parameters $FC^{\zeta}_{\xi,i,h}$ and $VC^{\zeta}_{\xi,i,h}$ correspond to fixed and variable costs (respectively) for state change processes at site i (state ξ to state ζ) during period h.

$$VC_{state} = \sum_{i\in I_e\cup I^{ini}_g\cup I_s\cup I_{hub}, h\in H} \left(VC^{liq}_{gas,i,h} q^{gas}_{1,i,h} + VC^{liq}_{sup,i,h} q^{sup}_{2,i,h} + VC^{gas}_{liq,i,h} q^{liq}_{1,i,h} \right.$$

$$\left. + VC^{sup}_{liq,i,h} q^{liq}_{2,i,h} + VC^{gas}_{sup,i,h} q^{sup}_{1,i,h} \right) \tag{74}$$

$$+ \sum_{i\in I^{sea}_g, h\in H} \left(FC^{dir}_{liq,i,h} y^{dir}_{i,h} + FC^{ind}_{liq,i,h} y^{ind}_{i,h} + \frac{VC^{dir}_{liq,i,h}}{\mu^{dir}_{liq,i}} b^{dir}_{i,h} + \frac{VC^{ind}_{liq,i,h}}{\mu^{ind}_{liq,i}} b^{ind}_{i,h} \right)$$

$$+ \sum_{i\in I_g, h\in H} \left(\frac{VC^{res}_{gas,i,h}}{\mu^{res}_{gas,i}} b^{gas}_{i,h} + \frac{VC^{res}_{sup,i,h}}{\mu^{res}_{sup,i}} b^{sup}_{i,h} \right)$$

4.3.13. Linked site binaries

There are situations where the user would like to enforce that two given nodes have to be selected at the same time (Eq. 75), that the selection of one node implies the selection of the other (Eq. 76), or that at most one out of the two nodes considered can be selected over the whole time horizon (Eq. 77). Such constraints offer the flexibility to split a site into multiple subsites, as for stacked reservoirs (see Appendix B).

$$y_{i,h} = y_{i',h}, \forall (i,i') \in I_{eq}, h \in H \tag{75}$$

$$y_{i,h} \le y_{i',h}, \forall (i,i') \in I_{leq}, h \in H \tag{76}$$

$$\sum_{h\in H} \left(y_{i,h} + y_{i',h} \right) \le 1, \forall (i,i') \in I_{or} \tag{77}$$

5. Heuristic algorithm

Poor solution quality and long calculation times can hinder the effective use of optimization models to inform CCUS strategic planning. As the scope of studies and the number of factors to be considered grow (Whitman et al., 2022; Homsi et al., 2025), the likelihood of computational inefficiency increases. Although these issues have been investigated for pipeline-based CCUS (Whitman et al., 2022; Homsi et al., 2025), they remain unaddressed for multiperiod CCUS networks with multimodal transport.

Incorporating vehicle transportation, fleet sizing and fleet mix considerations to pipeline-based models further hampers optimization solver performance. For the model proposed in this study (Section 4), the vehicle part can add a considerable number of equations ($3|K_V||H| + 2|I_V||H| + 3|V||H|$), which can slow down solver progress, e.g. calculations of lower bounds on the objective function and branching at search tree nodes. For the case study presented in the next section (Section 6), this corresponds to roughly 80k additional equations. While algorithmic developments are not the prime focus of this work, we propose a simple heuristic procedure as a first step to address computational efficiency. This method is based on the observation that vehicle loads can serve as a proxy to the number of fleet trips. In general, this leads to an underestimation of the number of trips and vehicle costs since arc traversals with part loads (or empty loads) still count as full trips. However, this results in a considerable reduction in the number of equations and integer variables (trips) for large instances. While this "load model" is easier to solve, numerical tests have shown that warm starting it with a feasible solution can lead to shorter calculation times and reaching better solutions. The proposed heuristic, detailed below, therefore solves a sequence of MILP models, with the solution of each initializing the next.

Henceforth, the MILP described in Section 4 is referred to as the "full model" and denoted as $\mathbb{P}$. The "load model" ($\mathbb{P}_{\text{load}}$) is derived from the full model by substituting the number of trips ($n^{o}_{k,v,h}$) by the number of loads ($l_{k,v,h}$), removing superfluous Eqs. (53-55) and introducing new variables ($\widetilde{\theta}_{v,h}$, $\widetilde{n}_{v,h}$, $\widetilde{n}^{\text{new}}_{v,h}$, $\widetilde{TC}$, $\widetilde{TC}_V$) in Eqs. 56-60, 65 and 68 in lieu of their non-tilded counterparts. The "fleet sizing" model ($\mathbb{P}_{\text{fleet}}$) corresponds to the fleet sizing Eqs. (52-60, 68).

By underestimating the number of trips, the load model gives a lower bound on the fleet size. The first step of the proposed algorithm involves solving a simplified version of the load model where we force all decisions for facility location and pipelines to be done in the first time period. This first step aims to quickly provide a warm start for the load model (step 2). Since the solution of the load model cannot be used directly to initialize the full model, an intermediate phase (step 3) uses the fleet sizing model to calculate the fleet size, fleet mix and number of trips corresponding to the solution of step 2. A feasible point for the full model is then obtained (step 4) via the union of the solutions from steps 2 and 3, replacing the approximate fleet sizing variables from step 2 by those from the fleet sizing model found in step 3. As an optional final refinement (step 5), the full model can be solved using the output of step 4 as a starting point. As highlighted in the next section, foregoing this last step can yield high quality results at a fraction of the time required for solve of the full model.

Algorithm: warm-start heuristic

Inputs: solver termination limits for calculation time (t_1, t_2, t_3, t_4) and optimality gap (g_1, g_2, g_3, g_4)

Step 1: Load model with facility location and pipeline network design restricted to first period

Solve $\mathbb{P}_{\text{load}}$ minimizing $\widetilde{TC}$ with (t_1, g_1) limits and $y_{i,h} = y^{p}_{k,c,h} = y^{e}_{i,m,h} = 0$, for h > 1.

Step 2: Load model

Solve $\mathbb{P}_{\text{load}}$ minimizing $\widetilde{TC}$ with (t_2, g_2) limits and warm start from solution of step 1.

Step 3: Fleet sizing model

Solve $\mathbb{P}_{\text{fleet}}$ minimizing TC_V with (t_3, g_3) limits and with $x_{k,v,h}$, $x^{\text{load}}_{i,v,h}$, $x^{\text{unload}}_{i,v,h}$ and $\theta_{i,v,h}$ fixed to their values coming from step 2.

Step 4: Solution to full model

Do the union of the solutions of steps 2 and 3, replacing the values of $\widetilde{\theta}_{v,h}$, $\widetilde{n}_{v,h}$, $\widetilde{n}^{\text{new}}_{v,h}$, $\widetilde{TC}$, $\widetilde{TC}_V$ from step 2 by the values found in step 3. Recompute the objective function value.

Step 5 (optional): Solution refinement via full model

Solve $\mathbb{P}$ minimizing TC with (t_4, g_4) limits and warm start from solution of step 4.

6. Illustrative case study

6.1. Context

Preliminary tests successfully validated outputs of the proposed model (up to 2 decimals) on six pipeline-only case studies from the SimCCS Gateway team (Pamidighantam et al., 2020). The present section focuses on demonstrating the features of the proposed model and heuristic algorithm (sections 4 and 5) on an illustrative Canadian case study. The region of interest covers Eastern Canada, spanning Ontario, Quebec and the Atlantic provinces. In this region, we consider all industrial facilities that emit at least 300 ktonnes of CO_2 annually. This represents total emissions of 57.6 Mtonnes/year from 144 emission points (flue gas stacks) located at 71 industrial facilities. For those locations, emitter data (Environment and Climate Change Canada, 2024) and MEA-based capture cost parameters were obtained following the methodology of Symonds et al. (2025). Three geological reservoirs are considered: the inland Cambrian sandstone aquifer located on the North shore of Lake Erie in Ontario (Baghmolaei et al., 2024), the inland Gentilly Block of the Potsdam sandstone in the St-Lawrence Platform of Quebec (Konstantinovskaya et al., 2023) and the Sable subbasin offshore of Nova Scotia (Carey et al., 2023). Storage costs were estimated using the functional forms of (McCollum and Ogden, 2006).

We focus on transportation via a combination of trains, ships and new pipelines (inland and offshore). As in Nie et al. (2021), six types of transport ships are considered, with capacities ranging from 6 to 50 ktonnes and five types of injection ships of capacities from 10 to 50 ktonnes. All data related to ships, intermediate floating platforms and auxiliary processes (liquefaction, pressurization, buffer storage) is taken from Nie et al. (2021). A total of 14 ports are considered for illustrative purposes (Figure 5). Rail transportation is done via trains of 100 railcars, each railcar having a capacity of 80 tonnes, using data from Han and Lee (2011). Pipeline costing is approximated via a cost function with two piecewise linear segments. Pipeline diameters are calculated following the approach of Vandeginste and Piessens (2008), while pipeline cost parameters are estimated using a weighted sum of cost per unit length and unit diameter values, based on the geographic features along their path. Different terrain conditions were considered using the factor weights of Morgan et al. (2023).

Fig. 5 shows the geographical scope of this case study. Potential pipeline routes (red) were generated by rasterizing the region of interest,

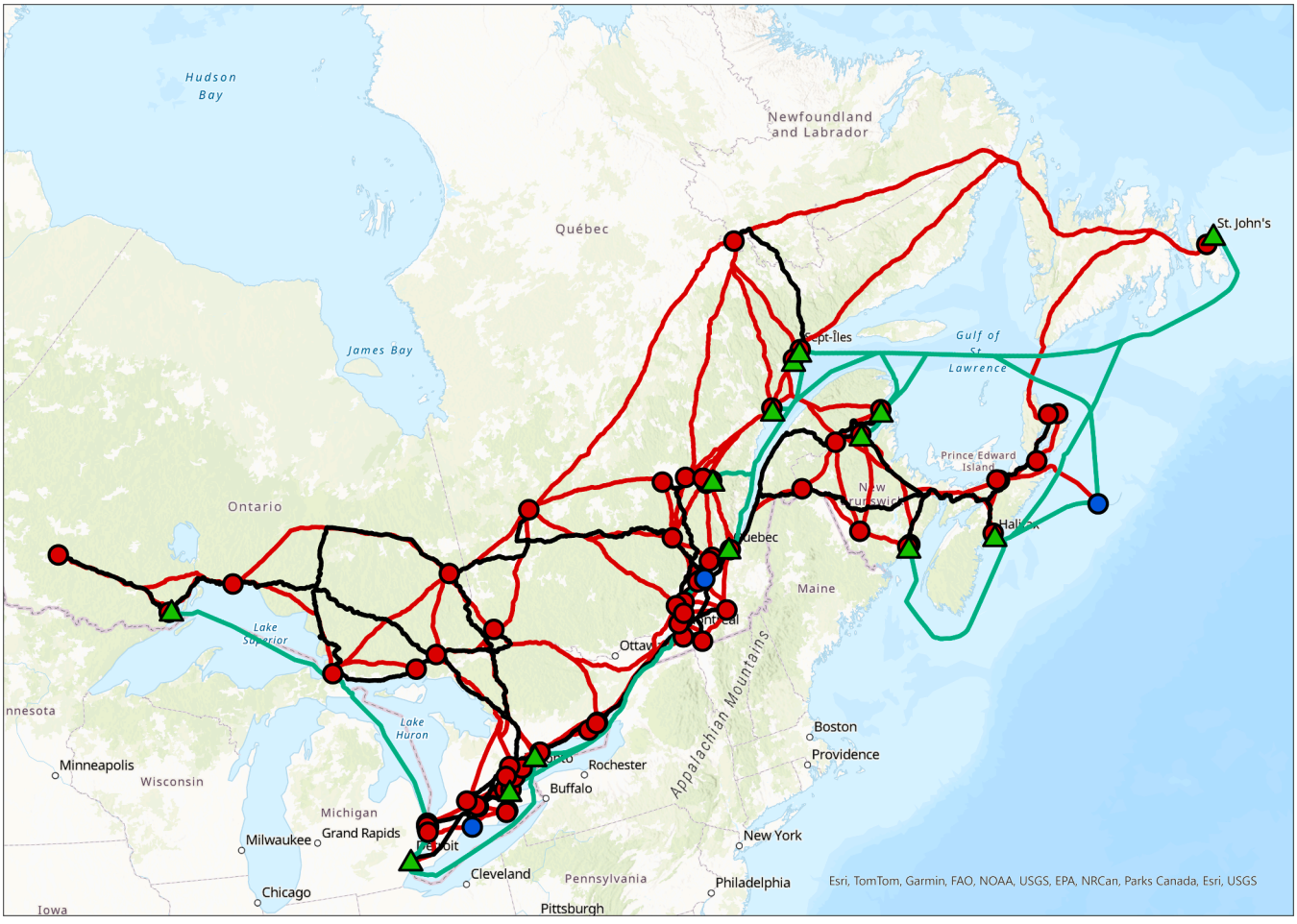


Fig. 5. Candidate transportation network: emitters (red dots), reservoirs (blue dots), ports (green triangles), candidate pipeline routes (red), railways (black), shipping lanes (green).

defining a cost map over each raster cell and calculating least cost paths for a Delaunay triangulation of the set of nodes, as in Hoover et al. (2020) and Morbee et al. (2012). This was implemented in the ArcGIS Pro software (Esri, 2025) taking into account the following geographical features: water bodies, wetlands, land slope, population density, protected/conserved areas, First Nations territories and existing rights-of-way (pipeline, rail and power transmission corridors). Each geographical factor's contribution to pipeline costing was weighed using the relative factors of Morgan et al. (2023) for regions designated as flat dry land, wetlands, rivers, high population, offshore and mountainous. These weighed costs were then summed to define the pipeline cost map and candidate network following the methodology of Hoover et al. (2020). It should be noted that these geographical weights come from a comparative study of pipeline capital costing across different regions in the United States (Morgan et al., 2023). As such, their application to the Canadian context is purely for illustrative purposes. Work is currently under way to update these values to the Canadian context. Therefore, an in-depth study of the impact of these factors on the candidate pipeline network topology and on pipeline costing will be the topic of a future study.

Shipping lanes (green) are based on data from Fisheries and Oceans Canada (2023) and U.S. Department of Transportation, Bureau of Transportation Statistics (2023). The Canadian railway system data from Natural Resources Canada (2021) was reduced to the rail network shown (in black) by considering a shortest path subnetwork linking the nodes of interest and by adding rail spurs to reach otherwise inaccessible sites.

While the proposed model can be used at the local scale (e.g. small industrial clusters), at the inter-provincial scale of the present study, rail spur costs are several orders of magnitude smaller than capture, storage and other transport costs. Therefore, they were neglected. Although the proposed model supports the inclusion of sites solely dedicated to transshipment (I_{hub}), establishing potential locations for such sites would require investigating space considerations for buffer storage tanks and state change processes (e.g. liquefaction), as well as taking into account land use regulations and other considerations which fall outside the scope of this work. Therefore, we assumed that transshipment is possible at emitter sites, reservoirs and ports via the potential addition of buffer storage capacity and auxiliary state change processes.

6.2. Results and discussion

Since reservoir characterizations and transport cost analyses are currently underway to gather data nationwide in Canada, the results presented below are intended solely for illustrative purposes. All monetary values reported are in 2025 Canadian dollars (CAD). Numerical optimization models and algorithms were implemented in GAMS 49.4.0 (GAMS Development Corporation, 2025) using CPLEX 22.1.2. All numerical experiments were conducted on a workstation with an Intel Xeon Platinum 8468 processor and 2 TB of memory.

Prior to optimization, a number of modelling decisions must be made, notably the choice of time discretization. As will be shown in the following analysis, the time partition chosen can significantly influence result quality (objective function value) and calculation times. Although the proposed model supports non-uniform time partitions, to simplify subsequent analysis, we assume that time periods are of equal duration.

We consider a planning horizon spanning 20 years, starting in 2025. We impose annual capture and storage targets (as lower bounds) that increase linearly and uniformly from one time period to the next – i.e., in increments of 100/N percent, where N denotes the number of time periods – reaching 100 % of annual emissions being captured and stored during the last period (Fig. 6).

6.2.1. Time discretization: impact on solver performance

We begin by comparing the performance of commercial algorithm CPLEX solving the full model, with that of the proposed heuristic, which uses CPLEX to solve a sequence of simplified model approximations (Section 5). All possible partitions of the 20-year time horizon in periods of equal duration are considered. All runs are done with a 1 day

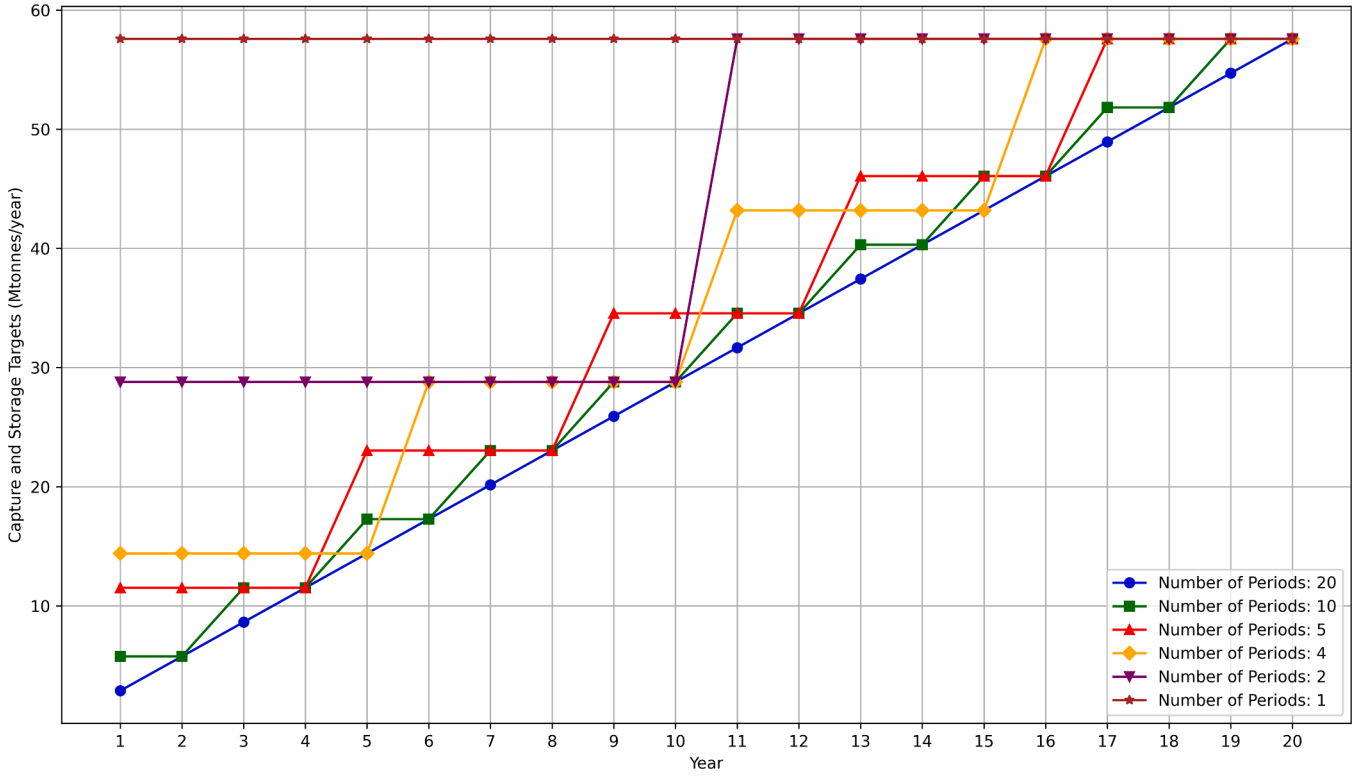


Fig. 6. Evolution of capture and storage targets as a function of number of periods.

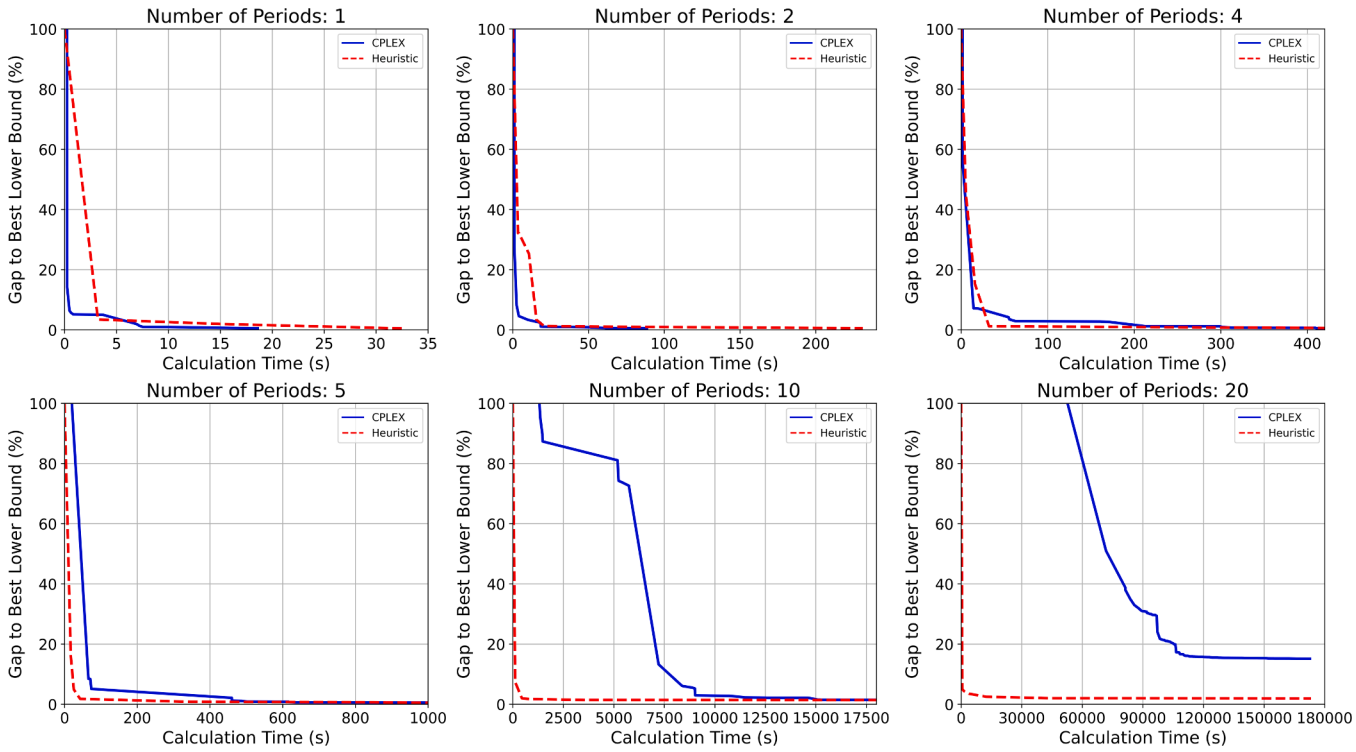


Fig. 7. CPLEX performance (solid line) and heuristic performance (dotted line) for 1, 2, 4, 5, 10 and 20 equipartitioned time periods.

calculation time limit, except for the 20 period case, where this limit is extended to 2 days. All CPLEX runs in standalone mode and for heuristic steps 2, 3 and 5 were carried out with an optimality gap limit of 0.5 %. For step 1 of the heuristic, a gap limit of 1 % was used.

For large instances of the proposed model (20+ periods), during preliminary tests, CPLEX standalone exhibited poor performance under default settings, sometimes failing to reach optimality gaps lower than 50 % after one day. For this reason, non-default CPLEX options were investigated. Instructing CPLEX to prioritize the search for hidden solutions (mipemphasis = 4) yielded the best performance. This option was also used to solve submodels of the heuristic.

For each planning horizon discretization, Fig. 7 shows how the performance of CPLEX standalone (solid line) and that of the heuristic (dotted line) vary with respect to calculation time. In order to have a uniform metric to compare results from both algorithms, we use the relative gap between the solution found (incumbent) and CPLEX's best lower bound [1]. The smaller this gap – i.e. the lower the performance curve – the closer the results are to optimality, with the caveat that the actual deviation from the optimum may be even smaller, since the estimated lower bound may not be tight. Results indicate that for smaller instances (1 or 2 periods), a general purpose solver like CPLEX can outperform the proposed heuristic, albeit by a small margin. For medium sized instances (4 or 5 periods), the heuristic tends to find better results slightly faster. As the number of time periods grows, the heuristic drastically outperforms CPLEX standalone (10 or 20 periods). Note that the near-horizontal trend at the tail of heuristic performance curves corresponds to the *optional* solution refinement step (Section 5), which lowers the total cost by less than 0.5 % for all multiperiod instances and by 1.45 % for the single period case. Therefore, if proving optimality is less important than obtaining high quality solutions rapidly, then this last step of the heuristic can be skipped. Appendix D contains further performance data.

These results highlight the fact that for larger instances, standalone commercial solvers may fail to find practical solutions for multiperiod models with multimodal transportation within days of calculations. In contexts where computing time budgets are limited, heuristics like the one proposed in this study are a computationally efficient alternative.

6.2.2. *Time discretization: impact on costs*

In this subsection, we investigate the costs associated to the best results found for each possible time discretization. Recall that we assume linear and uniformly increasing capture/storage targets (Fig. 6) and time periods of equal duration. It follows that increasing the number of periods yields a decrease in the total captured and stored amounts of CO_2, i.e. in Fig. 6, the (discrete) area under the curves decreases when more time periods are considered. Expectedly, the total cost is therefore a decreasing function of the number of periods. More precisely, the 1, 2, 4, 5 and 10 period cases lead to differences in total cost of 188 %, 69 %, 28 %, 21 % and 5 % (respectively) relative to the 20 period case. To provide a more meaningful basis for comparison, we consider the total cost *per tonne* of CO_2 (Fig. 8).

From this standpoint, for the 1, 2, 4, 5 and 10 period cases we observe a relative difference in total cost per tonne of 51 %, 18 %, 7 %, 5 % and 0.4 % (respectively) compared to the 20 period case. For this metric, intermediate cases (e.g. 4, 5, 10 periods) can offer a compromise between cost estimate quality and computing time, depending on the use case.

However, considering more time periods brings the added value of illustrating an approximate infrastructure construction timeline. Moreover, by impacting the ramp-up rate of capture and storage targets, the choice of time partition affects the flows of CO_2 transported and can therefore influence which transport modes and routes are considered, depending on cost vs volume vs distance trade-offs. For example, for this case study the instances with 1 and 2 periods solely involve offshore pipelines to reach the Sable subbasin due to the large volumes involves. From 4 time periods onwards, ships are selected during initial ramp-up stages since the transported volumes are smaller.

[1] Gap to best lower bound = 100*|objective value of solution – best lower bound|/best lower bound, where the best lower bound is obtained by CPLEX standalone on the full model after hitting the imposed time limit.

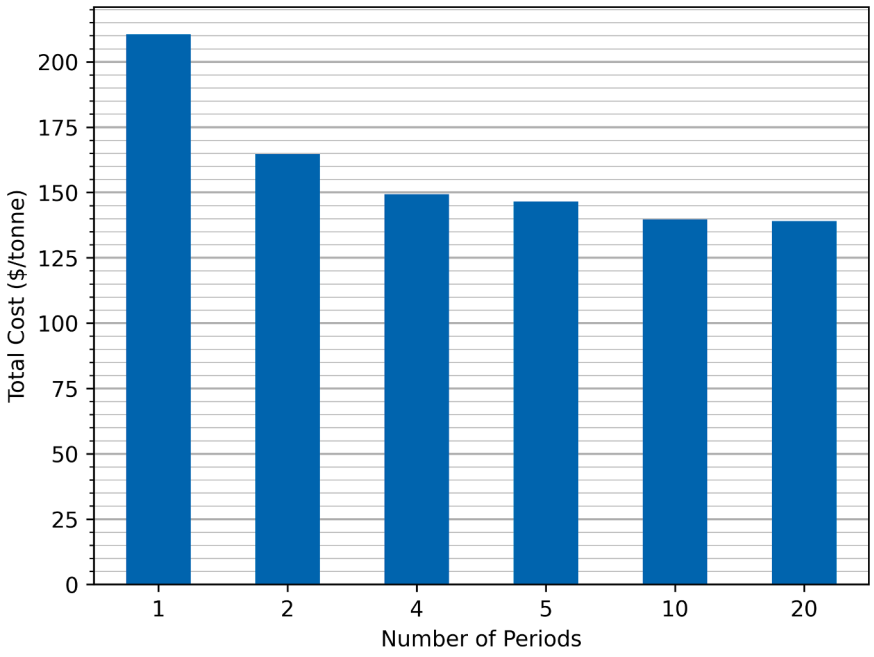


Fig. 8. Total cost per tonne cumulative over the whole time horizon for different temporal partitions.

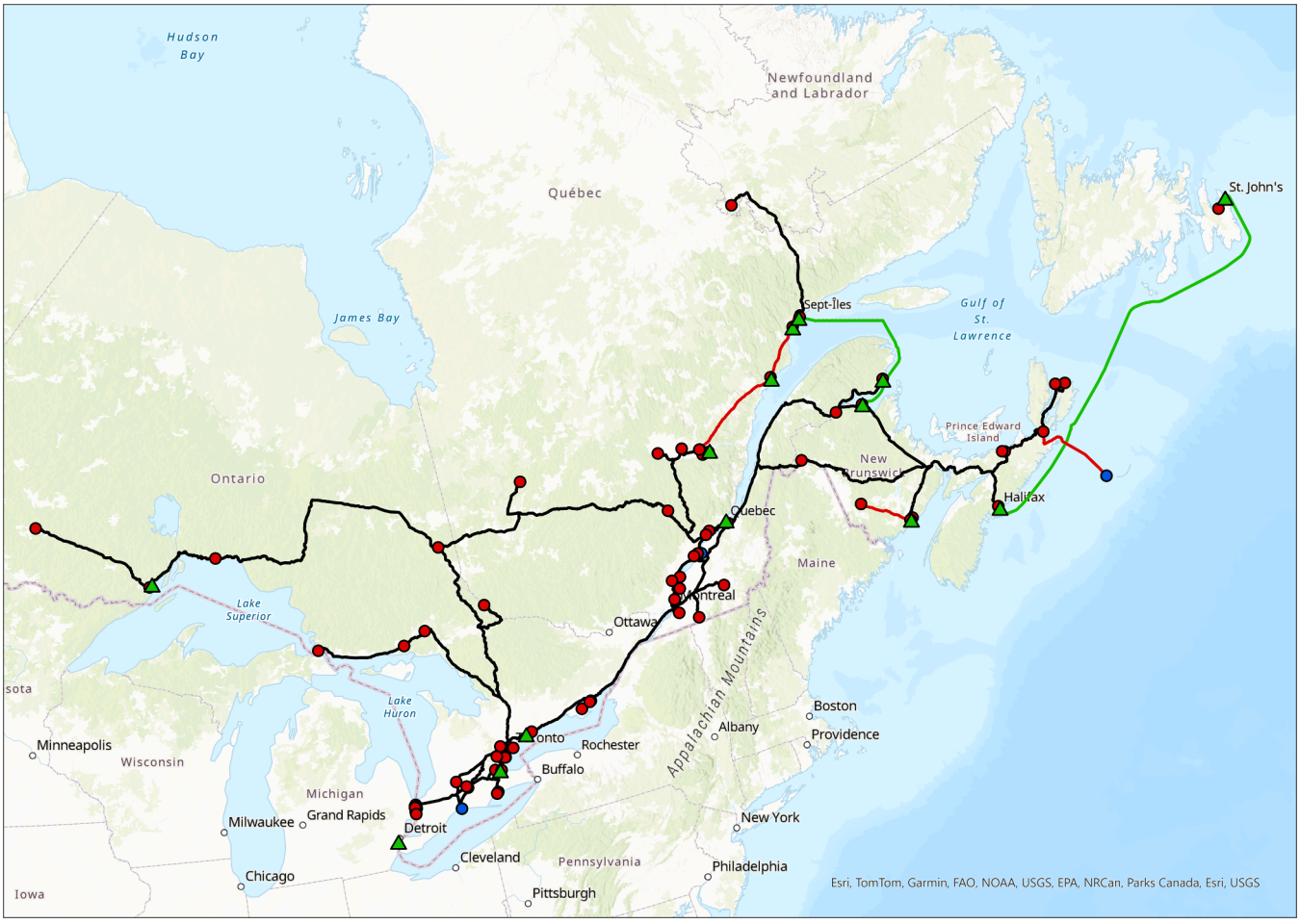


Fig. 9. CCUS value chain at last year of time horizon with 20 single-year periods assuming unrestricted transshipment at emitters, ports and reservoirs. Emitters (red dots), reservoirs (blue dots), ports (green triangles), pipelines (red), railways (black), shipping lanes (green).

In general, for a given distance, shipping costs need to be compared to pipeline economies of scale depending on volume. In that case, ships are favored for smaller volumes while offshore pipelines tend to be less costly for larger volumes. On the other hand, for a given CO_2 volume to be transported, the specific cost ($/tonne) of shipping is higher than that of offshore pipelines for short distances (IEAGHG, 2020). This is due in part to the fixed costs for vessels, liquefaction processes, temporary storage (tanks), as well as for loading and unloading equipment. When distances increase, ship fuel costs increase more slowly than offshore pipeline capital costs, up to a point where shipping becomes more economically viable than offshore pipelines past a certain distance.

An exhaustive investigation of the influence of the temporal resolution on the CCUS value chain topology and time-dependent constraints (e.g. site availability) falls outside the scope of this work. It will be the topic of a future study.

6.2.3. *Strategic plan over 20 years*

After 20 years, Fig. 9 shows the CCUS value chain yielded by the optimizer (20 periods). By the last year of the planning horizon, 49.4 Mtonnes/year are injected offshore (Sable subbasin) while the remainder of the storage target (8.2 Mtonnes/year) is met at inland reservoirs.

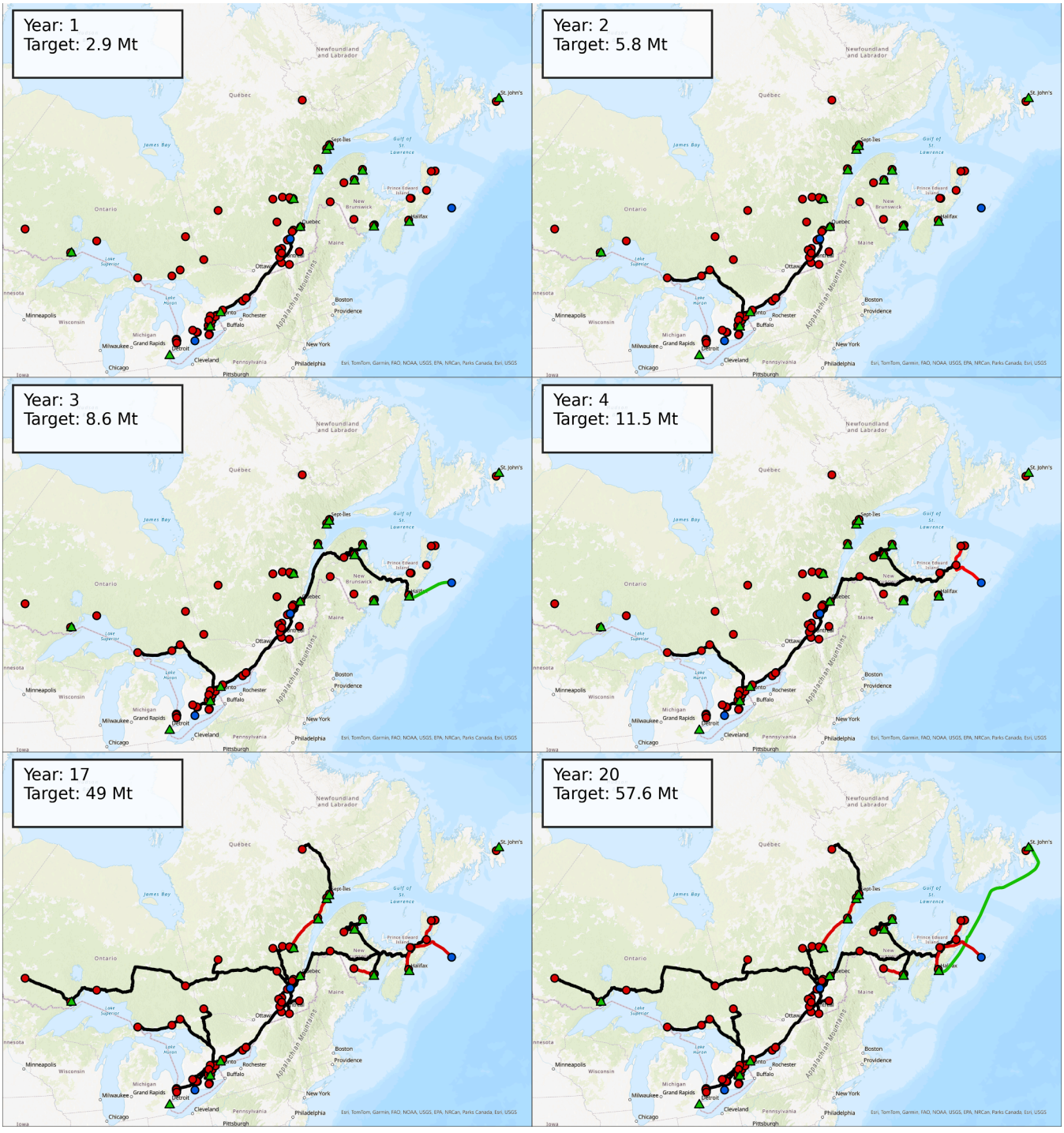


Fig. 10. Temporal snapshots of the CCUS value chain assuming restricted transshipment at emitters. Emitters (red dots), reservoirs (blue dots), ports (green triangles), pipelines (red), railways (black), shipping lanes (green).

The entirety of the flow along the offshore pipeline (49.4 Mtonnes/year) originates from trains unloading at the adjacent emitter site (Point Tupper). Recall that for all results shown until this point, no restrictions were put on intermodal transfers at emitters, ports and reservoirs. Since this is not always possible due to space limitations and other considerations, we now show how adding limits on transloading affects the CCUS value chain.

In order to implicitly constrain transshipment at emitting facilities, that is, to cap the capacity of buffer storage and state change processes, as well as the number of vehicle (un)loading trips, we limit the vehicle (un)loading rates at those sites to a maximum of 5 Mtonnes/year. This bound corresponds to admitting at most 4 train cargo discharges per day at those sites. As a result, the aforementioned situation where all offshore pipeline flow to the Sable subbasin comes from trains unloading at Point Tupper (Fig. 9) is avoided. To see this, we examine the temporal evolution of the CCUS value chain when this transshipment limit is added (Fig. 10).

During the first year, emitters in Ontario and Quebec along the Great Lakes and the Saint-Lawrence corridor capture 2.9 Mtonnes of CO_2, which is transported via 6 trains to the Gentilly Block reservoir (Quebec). Over the second year, more emitters are added to meet the twofold increase in capture target and as a result, the number of trains is doubled. At that time, the Gentilly Block reaches 97 % of its maximum annual injection rate. By the third year, this reservoir reaches its maximum annual injection rate and maintains it henceforth. To compensate, the other inland reservoir located on the Northern shore of Lake Erie (Ontario) starts operating at maximum injection rate from year 3 onwards. Since both inland reservoirs cannot meet the storage target during the third year due to insufficient injectivity, a single injection ship (10 ktonnes capacity) makes 46 trips to transport 0.457 Mtonnes of CO_2 from Nova Scotia to the Sable subbasin, whose injectivity is significantly larger than that of its inland counterparts.

With linearly increasing capture and storage targets (Fig. 6), the Sable subbasin makes up for the lack of inland injectivity (Fig. 11). In particular, by year 4, the increase in offshore transport volumes and the short distance involved (~140 km) make offshore pipelines more financially attractive than shipping for offshore storage. As a result, at the beginning of the fourth year an offshore pipeline with a capacity of 29.3 Mtonnes/year is introduced and by year 14, an extra 20.2 Mtonnes/year of capacity is added (parallel pipeline). From year 4 onwards, the transport network extends to reach more remote emitters, in Labrador City (at year 17) and Newfoundland (at year 19) - the latter requiring a single transport ship (40 ktonnes capacity) bringing 779 ktonnes of CO_2 yearly from Newfoundland to Nova Scotia for years 19 and 20.

Reservoir capacity and injectivity estimates can vary greatly depending on the data available and on the estimation methods used. For example, dynamic assessments of capacity can be up to 20 times inferior to those based on static analysis that ignore restrictions on injectivity (Lane et al., 2021). Low, mid and high permeability estimates can lead to similar variations in injectivity. A detailed and exhaustive evaluation of strategic plan robustness will be the subject of a future study.

With trains being the predominant transport mode for this scenario, Fig. 12 shows the evolution of the train fleet size and the total utilized railway length. We observe an increase of 6.3 trains per year on average (standard deviation 2.4) and by the end of the planning horizon, a total of 126 trains are used. For the shipping fleet, only the aforementioned two ships are used.

The total pipeline network length is 1253 km (Fig. 10). Since there is no available rail network linking emitters in Northeastern Quebec, a pipeline is built (~400 km). For the same reason, a pipeline (~130 km) links emitters in New Brunswick. The inland pipelines in Nova Scotia (~446km) are present due to the transshipment limits imposed at emitter sites (compare to Fig. 9, where no such limits exist). The aforementioned two parallel offshore pipelines (~140 km each) bring CO_2 to the Sable subbasin reservoir.

The majority of this strategic plan's total cost accumulated over the whole time horizon comes from capture operations (see Fig. 13). In the optimized plan, pipelines represent roughly one fifth of the total cost (19.1 %), while vehicles account for 4.3 %. This is due to the weight factors of (Morgan et al., 2023), where a 14:1 ratio is assumed between the cost (per unit length per unit diameter) of building pipelines on the seabed floor versus on flat-dry land.

In fact, the cost for the aforementioned parallel offshore pipelines corresponds to 85 % of the total pipeline cost. Reducing this offshore-to-inland pipeline cost ratio to 2:1 causes pipeline costs to drastically decline to only 6.1 % of total costs, which is slightly under the 6.2 % associated with storage. Recall that these weights are used to factor in

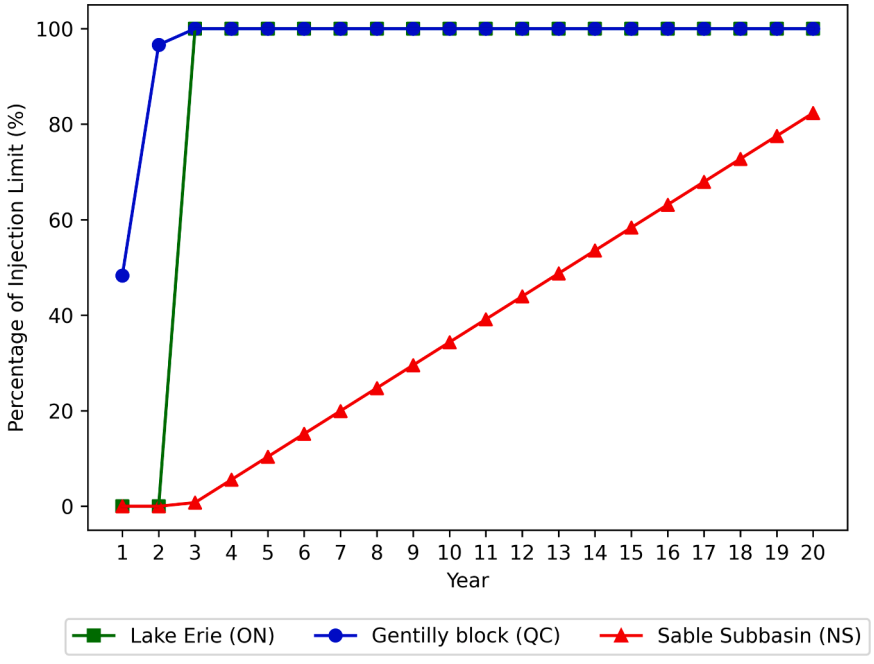


Fig. 11. Reservoir utilization per year for 20 period case with restricted transshipment.

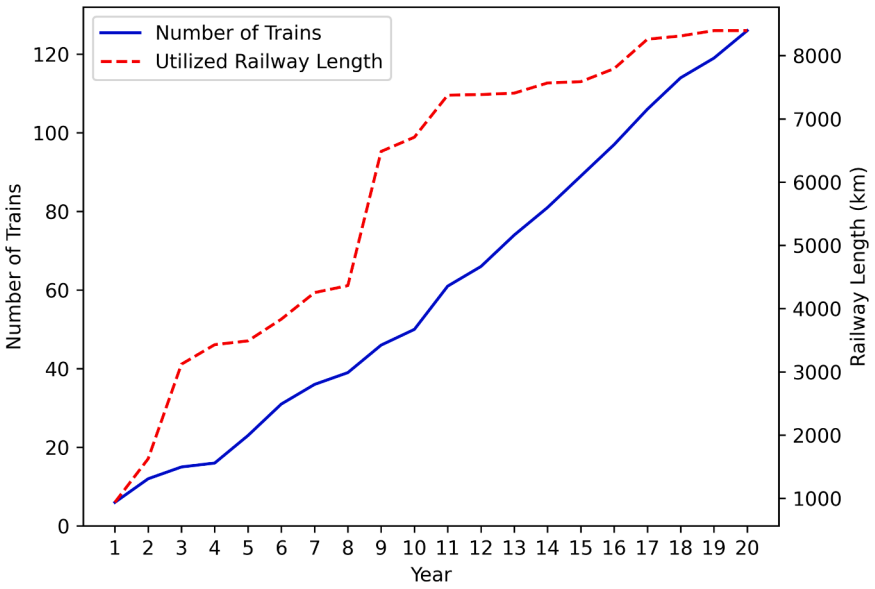


Fig. 12. Train fleet size and total utilized railway length per year for 20 period case with restricted transshipment.

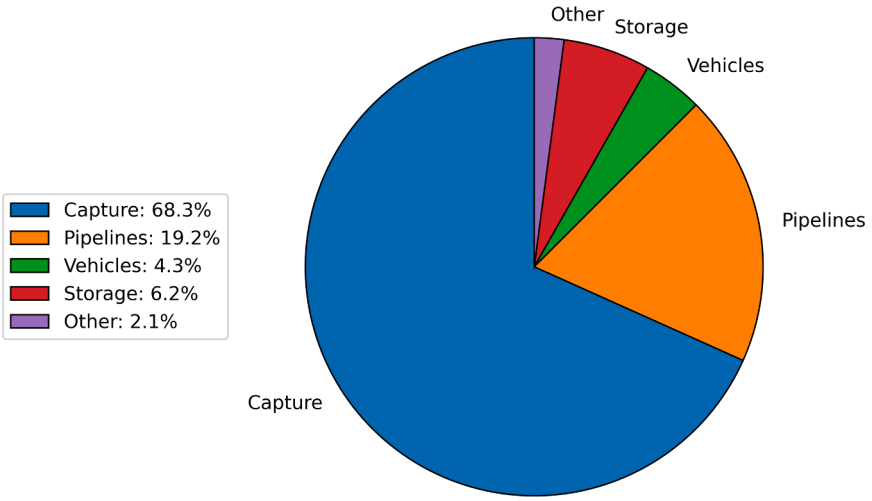


Fig. 13. Cost breakdown for the 20-period case with restricted transshipment. Other: state change processes and buffer storage.

the impact of various geographical features on pipeline costing and on the rasterized cost map (Section 6.1). As a consequence, they can influence both pipeline costs and the candidate pipeline network. This can in turn affect the combination of transport modes selected. Updating the US-based Morgan et al. (2023) weights to the Canadian context is the topic of an ongoing study. Therefore, a thorough sensitivity study of the influence of these weights on optimization results is the topic of a future investigation.

6. Conclusion

In this work, a novel model for the multiperiod strategic planning of CCUS with geographically explicit multimodal transportation was presented. In contrast with other approaches, this model features geographically explicit pipeline network routing - new and repurposed - coupled with ship, rail and truck transportation generalizing other models present in the literature. In particular, it covers the auxiliary processes and buffer storage necessary for transshipment, as well as fleet sizing and fleet mix considerations (new and existing vehicles). For offshore geological storage, the proposed model includes options for direct injection via ships, indirect injection via intermediate floating platforms and offshore pipelines. It also allows for multiple point emissions at the same industrial site and supports stacked reservoirs.

The new model's features were demonstrated on an illustrative Canadian case study covering Ontario, Quebec and the Atlantic provinces. As the results shown were for model and algorithmic demonstrative purposes only, this case study will be revisited in subsequent work, namely in light of ongoing efforts to refine nationwide capture, transport and storage data. Given the increasing geographical scope and the growing number of factors considered in large-scale CCUS assessments, end-users of such models must weigh the trade-off between the temporal resolution of the strategic plan, the quality of cost estimates obtained and computational performance. This is of particular relevance in use cases involving sensitivity analysis. The authors shed light on these interdependencies, which are seldom made explicit in the CCUS literature.

Although not the main focus of this work, a simple heuristic was introduced to reduce calculation times while improving result quality. For large instances, this new method was shown to yield significantly better results than leading commercial solver CPLEX (e.g. 10 % lower costs) at a fraction of the computing time (0.4 %), resulting in roughly 10 minutes of computation instead of 2 days. The authors expect that further improvements could be made by coupling leading algorithms for

multiperiod network design, such as pipeline slope scaling (Homsi et al., 2025), with the model approximations proposed in this work.

For this demonstrative case study, the costs and temporal evolution of CCUS infrastructure were analyzed within a 20 year planning horizon. The effects of reservoir injection rate and capacity limits, as well as transshipment restrictions were demonstrated, specifically on the overall value chain layout and on the transport modes selected. The authors also observed that different pipeline weighting factors associated to terrain characteristics can have an important impact on costs. They may also alter the mix of transport modes selected. Improving capture, transport and storage data accuracy across Canada is the subject of ongoing work. Although the proposed model can be used to select among different locations for new sites dedicated to transshipment, identifying such facilities requires analyzing multiple factors, such as space requirements and land use regulations. This topic and the inclusion of other constraints that could limit transit along railways and shipping lanes are relegated to future work.

The strategic planning of CCUS requires the careful management of multiple sources of uncertainty, notably those related to reservoir characterization. As other studies have shown, overlooking those factors can lead to unplanned costs, unplanned emissions and hinder the feasibility of capture targets (Middleton and Yaw, 2018). Addressing these considerations in a multiperiod context with geographically explicit multimodal transport remains an open question.

CRediT authorship contribution statement

Etienne Ayotte-Sauvé: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Robert Yandon:** Software, Resources, Data curation. **Philippe Navarri:** Writing – review & editing, Supervision, Resources, Project administration, Conceptualization. **Robert Symonds:** Writing – review & editing, Supervision, Resources, Project administration, Conceptualization. **Robin Hughes:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition, Conceptualization. **Marzieh Shokrollahi:** Writing – review & editing, Resources. **Rebecca Modler:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Nomenclature

Acronyms

CCUS	carbon dioxide capture utilization and storage
FSI	floating storage and injection platform
GIS	geographical information software
MILP	mixed integer linear programming

Sets

H: Set of planning time periods $h = 1,\ 2, \ldots, |H|$.

H_0: Set of extended time periods, i.e. $H_0 = H \cup \{0\}$. Planning time periods start at $h = 1$. The period $h = 0$ is only used to define conditions at the beginning of the planning time horizon.

I: Set of nodes, i.e. $I = I_e \cup I_g \cup I_s \cup I_{mix} \cup I_{hub}$

I_e: Set of source nodes (emitters)

I_g: Set of geological reservoir nodes ($I_g = I_g^{inl} \cup I_g^{sea}$)

I_g^{inl}: Set of inland geological reservoir nodes ($I_g^{inl} \subseteq I_g$)

I_g^{sea}: Set of offshore geological reservoir nodes ($I_g^{sea} \subseteq I_g$)

I_s: Set of port nodes

I_{hub}: Set of inland transshipment nodes

I_{mix}: Set of junction nodes (i.e. a mixer followed by a splitter) for pipelines (gas, supercritical), shipping routes, railroads and roads

V: Set of all vehicle types ($V = V_r \cup V_s \cup V_t$, disjoint union). A vehicle type is characterized by its cargo capacity, speed, fuel consumption, variable and fixed costs, loading and unloading rates and berth time (ships). See parameters section.

V_r: Set of all train types.

V_s: Set of all ship types.

V_s^{ind}: Set of all transport ship types (no direct injection equipment on board).

V_s^{dir}: Set of all injection ship types (direct injection equipment on board).

V_t: Set of all truck types.

L: Set of (i,v) in $I \times V$ such that vehicles of type v in V can be loaded at site i in $I_e \cup I_g^{inl} \cup I_{hub} \cup I_s$

U: Set of (i,v) in $I \times V$ such that vehicles of type v in V can unload at site i in $I_e \cup I_g \cup I_{hub} \cup I_s$

I_V: Set of (i,v) in $I \times V$ such that node i has at least one inlet or outlet arc permitting flow of vehicles of type v

C: Set of pipeline capacity trends, i.e. line segments in piecewise linear M$ vs Mtonnes/year cost functions for pipeline arcs

K: Set of arc labels for gas pipelines, supercritical pipelines, shipping routes, railroads and roads.

K^{rt}: Set of arc labels for new railroads and new roads.

K_C: Set of (k,c) in $K \times C$ such that pipeline arc k admits capacity trend c, i.e. piecewise linear segment c

K_V: Set of (k,v) in $K \times V$ such that arc k admits vehicles of type v (shipping routes, roads or railroads)

N_{gas}: Set of (k,i,j,c) in $K \times I \times I \times C$ such that gas pipeline arc k going from i to j admits capacity trend c

N_{sup}: Set of (k,i,j,c) in $K \times I \times I \times C$ such that supercritical pipeline arc k going from i to j admits capacity trend c

N_V: Set of (k,i,j,v) in $K \times I \times I \times V$ such that arc k going from i to j admits vehicles of type v (shipping routes, roads or railroads)

M: Set of CO_2 capture units. The term "capture unit" refers to "emitting process + capture + purification + compression + drying" I_M: Set of tuples $(i,m) \in I_e \times M$ where capture unit m is located at emitter site i in I_e

I_M^{gas}: Set of tuples $(i,m) \in I_M$ for which m is a capture unit with gas CO_2 output

I_M^{sup}: Set of tuples $(i,m) \in I_M$ for which m is a capture unit with supercritical CO_2 output

I_M^{liq}: Set of tuples $(i,m) \in I_M$ for which m is a capture unit with liquid CO_2 output

I_{eq}: Set of tuples $(i,i') \in (I \backslash I_{mix}) \times (I \backslash I_{mix})$ for which $y_{i,h} = y_{i',h}, \forall h \in H$

I_{leq}: Set of tuples $(i,i') \in (I \backslash I_{mix}) \times (I \backslash I_{mix})$ for which $y_{i,h} \leq y_{i',h}, \forall h \in H$

I_{or}: Set of tuples $(i,i') \in (I \backslash I_{mix}) \times (I \backslash I_{mix})$ for which the sum of $y_{i,h} + y_{i',h}$ over all h in H is at most 1, i.e. at most one site between i and i' can be selected over the whole time horizon

Parameters

O_h: Duration of time period h (year)

$Q_{k,c}^{max}$: Maximal annual capacity of a pipeline with trend c (i.e. linear piece c) built on arc k (Mtonnes/year)

Q_k^{rt}: Maximal flow of CO_2 on new rail or new road k (Mtonnes/year)

$Q_{i,m,h}^{e}$: Maximal annual CO_2 capture capacity of unit m at emitter site i in Ie during period h (Mtonnes/year)

$Q_{i,h}^{g}$: Maximal annual injection rate (i.e. injectivity) of reservoir at geological site i during period h (Mtonnes/year)

Q_i^{res}: Lifetime capacity of reservoir site i (Mtonnes)

Q_i^{liq} : Maximal value for the sum of intermediate flows $x_{1,i,h}^{liq}$ and $x_{2,i,h}^{liq}$ at inland site I, big-M constant (Mtonnes/year)

Q_i^{gas}: Maximal value of intermediate flow $x_{1,i}^{gas}$ at site I, big-M constant (Mtonnes/year)

Q_i^{sup} : Maximal value of intermediate flow $x_{1,i}^{sup} + 1_{i \in I_g^{sea}} x_{2,i}^{sup}$ at site I, big-M constant (Mtonnes/year)

$Q_{i,v}^{load}$: Maximal tons per year of CO_2 loaded on all vehicles of type v at site i, big-M constant (Mtonnes/year)

$Q_{i,v,h}^{unload}$: Maximal tons per year of CO_2 unloaded from all vehicles of type v at site i during period h, big-M constant (Mtonnes/year)

$\widetilde{Q}_{i,v}^{load}$: Loading rate of a single vehicle of type v at site i, loading a single full cargo (Mtonnes/year)

$\widetilde{Q}_{i,v}^{unload}$: Unloading rate of a single vehicle of type v at site i, unloading a single full cargo (Mtonnes/year)

$\widetilde{Q}_{i,v,h}^{dir}$: Unloading rate of a single injection ship of type v for direct injection at site i during period h, unloading a single full cargo (Mtonnes/year)

$\widetilde{Q}_{i,v}^{ind}$: Unloading rate of a single ship of type v to an FSI at site I, unloading a single full cargo (Mtonnes/year)

Q_v^{cap} : Cargo capacity of a vehicle of type v (Mtonnes)

δ_k : Length of arc k (km)

$\sigma_{k,v}$: Average speed of a vehicle of type v on arc k (km/year)

τ_v^{max} : Maximal proportion of a year a vehicle of type v is active, i.e. 1 - proportion of a year this vehicle must be inactive, $\tau_v^{max} \in [0,1]$. For ships, berth time between trips is counted as active time (year/vehicle)

$\tau_{i,v}^{berth}$: Proportion of a year a ship of type v is berthed at site i before a single trip; $\tau_{i,v}^{berth} \in [0,1]$ This excludes loading-unloading and planned downtime in a year where ships must remain inactive (year/trip)

$\mu_{\xi,i}^{\zeta}$: For state change process from state ξ to state ζ, ratio of outlet CO_2 flow over inlet CO_2 flow. In the interval]0,1]. ξ= gas, liq, sup; ζ = gas, liq, sup, dir, ind, res (-)

$VC_{k,c,h}^{p}$: Variable cost for pipeline arc k of capacity c during period h ($/tonne)

$FC_{k,c,h}^{p}$: Fixed cost for pipeline arc k of capacity c during period h (M$)

$FC_{k,v,h}^{o}$: Fixed costs (OPEX and CAPEX) proportional to the annual number of vehicle trips of a given type v along a route k during period h (M $/trip)

$FC_{v,h}^{rst}$: Fixed cost proportional to the number of vehicles of type v during period h. This excludes fixed costs (e.g. capital) for new vehicles, see $FC_{v,h}^{\text{new}}$ (M$/vehicle)

$FC_{v,h}^{\text{new}}$: Fixed cost proportional to the number of new vehicles of type v purchased during period h (M$/vehicle)

$FC_{k,h}^{rt}$: Fixed cost for building a rail spur or road on arc k during period h (M$)

$VC_{k,v,h}^{rst}$: Variable costs (OPEX and CAPEX) proportional to the annual CO_2 tonnage transported by a fleet of vehicles of a given type v on a given route k during period h ($/tonne)

$VC_{v,h}^{time}$: Variable costs (OPEX and CAPEX) proportional to the cumulative period of activity (in years) of the fleet of vehicles of type v during period h (M$/year)

$VC_{i,v,h}^{\text{load}}$: Variable loading cost at site i for vehicles of type v during period h ($/tonne)

$VC_{i,v,h}^{\text{unload}}$: Variable unloading cost at site i for vehicles of type v during period h ($/tonne)

$FC_{i,m,h}^{e}$: Fixed cost for capture unit m at site i during period h (M$)

$VC_{i,m,h}^{e}$: Variable cost for capture unit m at site i during period h ($/tonne)

$FC_{i,h}^{g}$: Fixed cost for opening geological site i during period h (M$)

$VC_{i,h}^{g}$: Variable cost for injecting CO_2 from pipelines at geological site during period h ($/tonne)

$VC_{i,h}^{dir}$: Variable cost for direct injection from ships at offshore reservoir i during period h, excludes conditioning ($/tonne)

$VC^{ind}_{i,h}$: Variable cost for indirect injection from intermediate FSI at offshore reservoir i during period h, excludes conditioning ($/tonne)

$FC^{dir}_{i,h}$: Fixed cost specific to direct injection from ships at offshore reservoir i during period h, excludes conditioning (M$)

$FC^{ind}_{i,h}$: Fixed cost specific to indirect injection from FSI at offshore reservoir i (excluding FSI cost, see FC^{tank}_{i}) during period h, excludes conditioning (M$)

$FC^{\text{tank}}_{i,h}$: Fixed cost associated to CO_2 storage tank (or FSI) at site i during period h (M$)

$VC^{\text{tank}}_{i,h}$: Variable cost associated to CO_2 storage tank (or FSI) at site i during period h ($/tonne)

$FC^{\zeta}_{\xi,i,h}$: Fixed cost for state change at site i (state ξ to state ζ) during period h, ξ= gas, liq, sup; ζ = gas, liq, sup,dir, ind, res (M$)

$VC^{\zeta}_{\xi,i,h}$: Variable cost for state change at site i (state ξ to state ζ) during period h. Proportional to installed capacity available (Mtonnes/year), i.e. maximal input flow rate. ξ= gas, liq, sup; ζ = gas, liq, sup, dir, ind, res. ($/tonne)

$FC^{site}_{i,h}$: Aggregate of all fixed costs that multiply site binary (see variable y_i) during period h (M$)

$F_{i,0}$: Initial buffer storage or FSI level at node i (Mtonnes)

$\widehat{F}_{i,0}$: Initial buffer storage or FSI capacity at node I (Mtonnes)

$N_{v,0}$: Initial number of vehicles of type v available at the beginning of the time horizon (vehicles)

$\widehat{f}^{min}_{i}$: Minimal buffer storage or FSI capacity at node i (Mtonnes)

$\widehat{f}^{max}_{i}$: Maximal buffer storage or FSI capacity at node i (Mtonnes)

β_i : Buffer storage or FSI capacity factor at site i. That capacity at a given time period must be at least the maximum of inlet and outlet flow to buffer storage or FSI during that time period multiplied by β_i, e.g. over a duration of 2 days with a safety margin factor of 1.2, $\beta_i = 1.2 * 2 / 365$ (-)

Variables

Below, the index h refers to a time period in the set H (as opposed to H_0), unless explicitly stated otherwise.

Binary variables

$y_{i,h}$: Takes value 1 iff site i in Ie or Is or Ihub or Ig is opened during period h, i.e. it becomes available at the beginning of period h onwards

$y^{p}_{k,c,h}$: Takes value 1 iff a pipeline of capacity trend c is built on arc k during period h, i.e. that pipeline becomes available at the beginning of period h onwards.

$y^{rt}_{k,h}$: Takes value 1 iff a railroad or road is built on arc k during period h, i.e. that arc becomes available at the beginning of period h onwards.

$y^{e}_{i,m,h}$: Takes value 1 iff emitter site i in Ie is retrofitted with capture unit m in M during time period h, i.e. m becomes available at node i at the beginning of period h onwards.

$y^{ind}_{i,h}$: Takes value 1 iff offshore geological storage site i in I^{sea}_{g} starts using an intermediate FSI (i.e. indirect injection) during time period h, i.e. from the beginning of period h onwards. Ship-based injection can be either direct or indirect, but not both.

$y^{dir}_{i,h}$: Takes value 1 iff offshore geological storage site i in I^{sea}_{g} starts using direct injection from ships (i.e. no FSI) during time period h, i.e. from the beginning of period h onwards. Ship-based injection can be either direct or indirect, but not both.

Integer variables (non-negative)

$n^{o}_{k,v,h}$: Annual number of trips (outings) of vehicles of type v along arc k during period h, $(k,v) \in K_V$. Carrying empty loads still counts as trips; see Eq. linking trips and loads. Cumulative over all vehicles of type v. (trip/year)

$n_{v,h}$: Annual number of vehicles of type v in V used in period h. (vehicle/year)

$n^{\text{new}}_{v,h}$: Number of new vehicles of type v in V purchased in period h in $H_0 = H \cup \{0\}$. For $h > 0$, those vehicles are available from the beginning of period h onwards. For $h = 0$, $n^{\text{new}}_{v,0}$ corresponds to the number of vehicles of type v already available at the beginning of the time horizon. (vehicle)

Real variables (non-negative)

$q_{k,c,h}$: Capacity of a pipeline with trend c (i.e. linear piece c) built on arc k during period h, $(k,c) \in K_C$. That pipeline becomes available at the beginning of period h onwards. (Mtonnes/year)

$x^{p}_{k,c,h}$: Annual tons of CO_2 transported on arc k for pipeline capacity trend c during period h, i.e. $(k,c) \in K_C$ (Mtonnes/year)

$x_{k,v,h}$: Annual tons of CO_2 transported by the fleet of vehicles of type v on arc k during period h, $(k,v) \in K_V$. Cumulative over all vehicles of type v. (Mtonnes/year)

$l_{k,v,h}$: Annual number of loads (proportion of max cargo, adimensional) for the fleet of vehicles of type v flowing along arc k during period h, $(k, v) \in K_V$. Cumulative over all vehicles of type v. (1/year)

$x^{\text{load}}_{i,v,h}$: Annual tons of CO_2 loaded on all vehicles of type v at site i during period h, (i,v) in L. Cumulative over all vehicles of type v. (Mtonnes/year)

$x^{\text{unload}}_{i,v,h}$: Annual tons of CO_2 unloaded from all vehicles of type v at site i during period h, (i,v) in U. Cumulative over all vehicles of type v. (Mtonnes/year)

$x^{ind}_{i,v,h}$: Linear reformulation of the product $x^{\text{unload}}_{i,v,h} \sum_{\tau=1}^{h} y^{ind}_{i,\tau}$ for i in I^{sea}_{g}, (i,v) in U, h H. Cumulative over all ships of type v. (Mtonnes/year)

$\widehat{a}_{i,h}$: Tons per year of CO_2 captured at site i in Ie during period h. Cumulative over all capture units available at site i. (Mtonnes/year)

$a_{i,m,h}$: Tons per year of CO_2 coming from capture unit m at site i in Ie during period h. (Mtonnes/year)

$x^{gas}_{1,i,h}$: Intermediate flow at node i in $I_e \cup I_g \cup I_s \cup I_{hub}$ during period h, see flow diagrams (Mtonnes/year)

$x^{gas}_{2,i,h}$: Intermediate flow at node i in $I_e \cup I^{inl}_{g} \cup I_s \cup I_{hub}$ during period h, see flow diagrams (Mtonnes/year)

$x^{sup}_{1,i,h}$: Intermediate flow at node i in $I_e \cup I_g \cup I_s \cup I_{hub}$ during period h, see flow diagrams (Mtonnes/year)

$x^{sup}_{2,i,h}$: Intermediate flow at node i in $I_e \cup I^{inl}_{g} \cup I_s \cup I_{hub}$ during period h, see flow diagrams (Mtonnes/year)

$x^{sup}_{3,i,h}$: Intermediate flow at node i in $I_e \cup I^{inl}_{g} \cup I_s \cup I_{hub}$ during period h, see flow diagrams (Mtonnes/year)

$x^{liq}_{1,i,h}$: Intermediate flow at node i in $I_e \cup I^{inl}_{g} \cup I_s \cup I_{hub}$ during period h, see flow diagrams (Mtonnes/year)

$x^{liq}_{2,i,h}$: Intermediate flow at node i in $I_e \cup I^{inl}_{g} \cup I_s \cup I_{hub}$ during period h, see flow diagrams (Mtonnes/year)

$q^{gas}_{1,i,h}$: Phase change process capacity (max. inlet flow) at node i in $I_e \cup I^{inl}_{g} \cup I_s \cup I_{hub}$ installed at period h, see corresponding x variable in flow

diagrams. Available from period h onwards. (Mtonnes/year)

$q^{sup}_{1,i,h}$: Phase change process capacity (max. inlet flow) at node i in $I_e \cup I^{inl}_g \cup I_s \cup I_{hub}$ installed at period h, see corresponding x variable in flow diagrams. Available from period h onwards. (Mtonnes/year)

$q^{sup}_{2,i,h}$: Phase change process capacity (max. inlet flow) at node i in $I_e \cup I^{inl}_g \cup I_s \cup I_{hub}$ installed at period h, see corresponding x variable in flow diagrams. Available from period h onwards. (Mtonnes/year)

$q^{liq}_{1,i,h}$: Phase change process capacity (max. inlet flow) at node i in $I_e \cup I^{inl}_g \cup I_s \cup I_{hub}$ installed at period h, see corresponding x variable in flow diagrams. Available from period h onwards. (Mtonnes/year)

$q^{liq}_{2,i,h}$: Phase change process capacity (max. inlet flow) at node i in $I_e \cup I^{inl}_g \cup I_s \cup I_{hub}$ installed at period h, see corresponding x variable in flow diagrams. Available from period h onwards. (Mtonnes/year)

$f_{i,h}$ level of buffer storage (inland sites) or FSI (offshore storage) at node i in $I_e \cup I_g \cup I_s \cup I_{hub}$ at the end of period h in H_0 (Mtonnes)

$\widehat{f}_{i,h}$: capacity of buffer storage (inland sites) or FSI (offshore storage) at node i in $I_e \cup I_g \cup I_s \cup I_{hub}$ added at the beginning of period h in H_0 (Mtonnes)

$\widehat{b}_{i,h}$: Tons per year of CO_2 injected at geological site i in Ig during period h. Cumulative over all wells. Cumulative over all incoming pipelines and ships (see flow diagrams). (Mtonnes/year)

$b^s_{i,h}$: Annual number of tons of CO_2 injected at offshore reservoir i in I^{sea}_g during period h coming from ships, either via direct injection or via intermediate FSI. Cumulative over all ship types and wells available at site i. (Mtonnes/year)

$b^{dir}_{i,h}$: Annual number of tons of CO_2 directly injected from ships at offshore reservoir i in I^{sea}_g during period h. Cumulative over all ship types and wells available at site i. (Mtonnes/year)

$b^{ind}_{i,h}$: Annual number of tons of CO_2 indirectly injected from intermediate FSI at offshore reservoir i in I^{sea}_g during period h. Cumulative over all ship types and wells available at site i. (Mtonnes/year)

$b^p_{i,h}$: Annual number of tons of CO_2 injected at reservoir i in I_g coming from pipelines during period h. Cumulative over all pipeline types and wells available at site i. (Mtonnes/year)

$b^{gas}_{i,h}$: Tons per year of CO_2 from gas pipelines injected at geological site i in I_g during period h. Cumulative over all wells available at site i. (Mtonnes/year)

$b^{sup}_{i,h}$: Tons per year of CO_2 from supercritical pipelines injected at geological site i in I_g during period h. Cumulative over all wells available at site i (Mtonnes/year)

$\theta_{i,v,h}$: Annual cumulative duration (in years) that all ships of type v in V_s spend unloading at storage site i in I^{sea}_g during period h ($(i,v) \in U$), i.e. doing direct injection or unloading into a FSI. (year)

$\widehat{\theta}_{v,h}$: Annual cumulative duration of activity (in years) of all vehicles of type v in V during period h. (year)

$CapCO2_h$: Total annual CO_2 captured during period h. Cumulative over all capture units. (Mtonnes/year)

$InjCO2_h$: Total annual CO_2 injected in geological sites during period h. Cumulative over all reservoir subsites. (Mtonnes/year)

Appendix B. Reservoir subsites

This section gives an example of how to use the different node types and the linked site binaries feature to define an inland geological storage site with subsites, i.e. a stacked reservoir. There are various ways to do this, depending on the site specifics. The model offers flexibility to decompose such reservoir nodes in a manner that reflects site characteristics.

In this example, the whole site is represented by a transshipment hub node and one inland geological storage node per subsite (see diagram below). Incoming vehicle and pipeline flows to the site are fed to the transshipment hub and corresponding output flows are split to feed each inland geological storage node.(Fig. B.1).

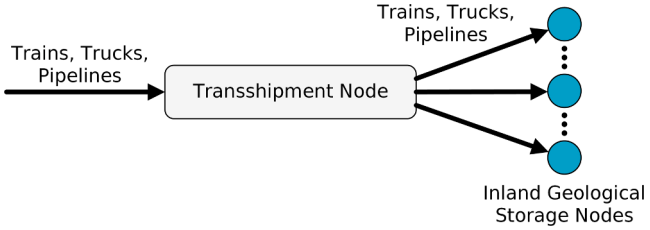


Fig. B.1. Decomposing an inland reservoir site into multiple subsites.

We assume that loading and unloading of vehicles into buffer storage occurs solely at the transshipment site, as well as all the state change processes. Therefore, at an inland geological storage site fed by this transshipment hub, only the "state change + injection" box of its flow diagram is used (see Fig. 1). These assumptions are made solely for illustrative purposes. As previously mentioned, site specifics can be considered by using other node types, as well as through variable and parameter assignations (see example below).

In the preceding diagram, let *i'* denote the transshipment node. For each inland geological storage node *i* in that diagram (corresponding to a storage subsite) and each time period *h*, we enforce the following assignations.

- Add (i, i') to the set I_{leq}, which will ensure that $y_{i,h} \leq y_{i',h}$, i.e., if the hub site is closed, then all subsites are closed.
- Set $\widehat{f}_{i,h} = 0$, i.e., set the upper bound to zero.
- Set $FC^{tank}_{i,h} = 0$ and $VC^{tank}_{i,h} = 0$.
- Set $x^{load}_{i,v,h} = 0$ for all $v \in V$ with $(i,v) \in L$, i.e., set the upper bound to zero.

- Set $\widetilde{Q}^{load}_{i,v} = 1$ for all $v \in V$ with $(i,v) \in L$, see fleet sizing Eqs..
- Set $VC^{load}_{i,v,h} = 0$ for all $v \in V$ with $(i,v) \in L$.
- Set $x^{unload}_{i,v,h} = 0$ for all $v \in V$ with $(i,v) \in U$, i.e., set the upper bound to zero.
- Set $\widetilde{Q}^{unload}_{i,v,h} = 1$ for all $v \in V$ with $(i,v) \in U$, see fleet sizing Eqs..
- Set $VC^{unload}_{i,v,h} = 0$ for all $v \in V$ with $(i,v) \in U$.
- Set $x^{gas}_{1,i,h} = x^{liq}_{1,i,h} = x^{liq}_{2,i,h} = x^{sup}_{1,i,h} = x^{sup}_{2,i,h} = 0$, i.e., set the upper bound to zero.
- Set $\mu^{liq}_{gas,i} = \mu^{liq}_{sup,i} = \mu^{gas}_{liq,i} = \mu^{sup}_{liq,i} = \mu^{gas}_{sup,i} = 1$.
- Set $VC^{liq}_{gas,i,h} = VC^{liq}_{sup,i,h} = VC^{gas}_{liq,i,h} = VC^{sup}_{liq,i,h} = VC^{gas}_{sup,i,h} = 0$.
- Set $FC^{liq}_{gas,i,h} = FC^{liq}_{sup,i,h} = FC^{gas}_{liq,i,h} = FC^{sup}_{liq,i,h} = FC^{gas}_{sup,i,h} = 0$.

Vehicle and pipeline costs associated to CO_2 transportation from the transshipment node to the associated storage subsites can then be defined. If these costs are to be neglected, then the following parameter assignations are made.

- Set $FC^{p}_{k,c,h} = VC^{p}_{k,c,h} = 0$ for all $(k,c) \in K_C$ such that k is a pipeline arc linking the hub to its associated storage sites.
- Set $FC^{o}_{k,v,h} = VC^{rst}_{k,v,h} = 0$ for all $(k,v) \in K_V$ such that k is a truck or rail arc linking the hub to its associated storage sites.
- Set $\delta_k = 0$ for all $k \in K$ such that k is an arc linking the hub to its associated storage sites.

Appendix C. Parameter definition formulas

This appendix lists relations used to define model parameters.

Site fixed costs

$$\mathrm{FC}^{site}_{i,h} = \begin{cases} FC^{load}_{i,h} + FC^{unload}_{i,h} + FC^{tank}_{i,h} + FC^{liq}_{gas,i,h} + FC^{liq}_{sup,i,h} + FC^{gas}_{liq,i,h} + FC^{sup}_{liq,i,h} + FC^{gas}_{sup,i,h}, & if\ i \in I_e \cup I_s \cup I_{hub} \\ FC^{g}_{i,h} + FC^{load}_{i,h} + FC^{unload}_{i,h} + FC^{tank}_{i,h} + FC^{liq}_{gas,i,h} + FC^{liq}_{sup,i,h} + FC^{gas}_{liq,i,h} + FC^{sup}_{liq,i,h} + FC^{gas}_{sup,i,h}, & if\ i \in I^{inl}_g \\ \qquad + FC^{res}_{gas,i,h} + FC^{res}_{sup,i,h} & \\ FC^{g}_{i,h} + FC^{load}_{i,h} + FC^{unload}_{i,h} + FC^{res}_{gas,i,h} + FC^{res}_{sup,i,h}, & if\ i \in I^{sea}_g \end{cases} \tag{C.1}$$

Appendix D. Computational performance: additional results

Table D.1 compares snapshots in time of CPLEX and the proposed heuristic for the Eastern Canada case study. For smaller instances (1 or 2 periods), we see that without the refinement step the heuristic can already get within 1.9 % of optimality in less than one minute. Adding this step enables it to come within a margin of 0.5 % of the estimated optimum in less than 5 minutes, which is slightly slower than CPLEX standalone. For 4 periods, the heuristic without a refinement phase reaches a gap to best bound of 0.9 % in less than 4 minutes, which is then narrowed to 0.5 % in less than 10 minutes by adding this final step. This is roughly 3 minutes slower than CPLEX standalone.

Table D.1
Heuristic (without refinement step) vs. CPLEX for 10 minute limit.

Number of periods	Heuristic 10 min.* limit		Heuristic 1 hour limit		CPLEX 10 min. limit		CPLEX 1 hour limit		CPLEX 1 day limit	
	Gap to bkb (%)**	Calc. time (s)	Gap to bkb (%)**	Calc. time (s)	Gap to bkb (%)**	Calc. time (s)	Gap to bkb (%)**	Calc. time (s)	Gap to bkb (%)**	Calc. time (s)
1	1.9 %	15	0.5 %	32	0.5 %	19	0.5 %	19	0.5 %	19
2	1.1 %	59	0.5 %	276	0.6 %	88	0.6 %	88	0.6 %	88
4	0.9 %	207	0.5 %	583	0.6 %	421	0.6 %	421	0.6 %	421
5	0.8 %	349	0.5 %	1620	0.8 %	600	0.5 %	3600	0.5 %	19740
10	2.0 %	449	1.4 %	3472	260 %	600	87.3 %	3600	1.3 %	85851
20	5.0 %	612	3.5 %	3810	90530 %	600	90530 %	3600	33.0 %	85728

*: Heuristic without its optional refinement step (step 5)
**: bkb: best known lower bound

As the heuristic is designed to tackle larger instances, it is unsurprising that it becomes more efficient than CPLEX standalone for 5, 10 and 20 periods, where the size of the vehicle submodel grows considerably. For 5 time periods, with runtime limits of 10 minutes and 1 hour, we see that the heuristic takes between 42 % and 55 % less time to reach the same gap as CPLEX. For 10 time periods, CPLEX standalone only reaches a gap to the best bound of 87.3 % after an hour and takes one day to lower that gap to 1.3 %. On the other hand, after less than 8 minutes the heuristic already reaches a gap to best bound of 2 % and further reduces it to 1.4 % within 58 minutes, without even requiring a refinement step (step 5). Adding the optional refinement phase at the end of the heuristic lowers that gap to 1.27 % after one day.

For 20 time periods, CPLEX standalone performance significantly worsens. It takes 15 hours to reach a gap to the best bound of 105 % and then reduces it to 33 % after one day. After 29.3 hours it gets to 19.9 % and settles at 15.1 % after two days. In sharp contrast, the heuristic gets within a margin of 5 % after only 10 minutes (without a refinement step). Letting it run for 64 minutes lowers this gap to 3.51 %, settling at 1.99 % after 1 day, after which steps 1-4 of the heuristic have reached optimality. Letting the final refinement phase (step 5) run for an additional day only lowers the total cost by 0.06 % and therefore brings little value.

Data availability

Data will be made available on request.

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